

A Geometric Observatory for Financial Regime Detection

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Abstract

We construct a seventeen-channel geometric feature panel derived from a spectral metric learning embedding (QCML) and evaluate it for financial regime detection across 17 crises spanning 2000–2024. Walk-forward nested HPO—our primary evaluation—yields unbiased out-of-sample Berry Phase Rate $d = 0.72$, winning 5/9 walk-forward crises; geometric methods produce 30% fewer false alarms than Random Forest (2.5 vs. 3.6/yr). In the offline event-study comparison (36 methods, 17 crises), Reduced State Purity achieves the highest median $d = 0.83$ (rank 1; conditional on bipartition choice), followed by Hamilton Markov-switching [22] ($d = 0.71$, rank 2), CUSUM ($d = 0.62$, rank 3), and Berry Phase Rate ($d = 0.61$, rank 4). The Friedman test ($\chi^2 = 250.6$, Iman-Davenport $p < 10^{-34}$) confirms significant ranking heterogeneity. Different geometric channels show per-crisis variation in performance across crisis types, though this pattern does not reach significance (permutation $p = 0.31$). On structurally novel crises, geometric observables maintain stable detection while supervised methods degrade (−25% for Random Forest, −58% for CUSUM). A pseudo-prospective holdout simulation (pipeline frozen at 2021-12-31) confirms generalization on three unseen post-2021 crises. All bootstrap CIs use circular block resampling [40] to account for serial correlation; Cliff’s delta is reported alongside Cohen’s d .

Keywords: spectral metric learning, regime detection, Berry curvature, quantum Fisher information, Fubini–Study metric, geometric observatory, financial crises, QCML.

1 Introduction

Detecting regime transitions—shifts between qualitatively different market states—is a central problem in quantitative finance. Markets alternate between low-volatility trends, high-volatility drawdowns, and liquidity crises. Classical approaches include Hidden Markov Models [4, 22], Bayesian Online Changepoint Detection [2], and supervised classifiers. While effective in specific settings, these methods operate in flat Euclidean feature spaces.

Quantum Cognition Machine Learning (QCML) [16]—introduced in industry white papers [34, 43] and subsequently validated in peer-reviewed work [16]—encodes data points as unit vectors in a finite-dimensional Hilbert space and represents features as Hermitian operators. The map $x \mapsto |\psi(x)\rangle$ from feature space to projective Hilbert space $\mathbb{CP}^{n-1}$ endows the data manifold with a Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. Candelori et al. [16] showed that this geometry robustly captures intrinsic dimension; Abanov et al. [1] developed the mathematical theory. Financial applications include forecasting [43], similarity learning [42, 44], and biomedical prediction [17].

Classical computation. All QCML computations—Hamiltonian construction, eigendecomposition, metric and curvature evaluation—run entirely on classical hardware. The “quantum” label refers to the mathematical formalism (Hilbert space, Hermitian operators, spectral decomposition), not to quantum computing resources.

Our contribution is a *systematic geometric feature panel*: seventeen detection channels extracted from the QCML embedding, each probing a distinct geometric property of the data manifold (Table 2). We observe that different channels show per-crisis variation in performance—volatility shocks, systemic contagion, exogenous events, slow burns, and liquidity breakdowns tend toward different geometric winners, though this pattern is suggestive rather than statistically significant on 17 crises (permutation test $p = 0.31$; Appendix M).

The evaluation is framed as *crisis-window separability*: given a score time series, we ask whether the distribution of scores during a known crisis window is statistically separated from the non-crisis distribution. This is an *offline event study*—it measures sensitivity to regime transitions, not real-time predictive power. A supplementary walk-forward benchmark (Section 5.5) tests causal detection with strictly past-fitted preprocessing.

Clarification on supervision level. Score construction is unsupervised: the QCML observables require no labeled crisis data to compute z-scores (Algorithm 1). However, the overall pipeline is *semi-supervised*: hyperparameters (Hilbert dimension, PCA components, operator method, rolling window) are selected via Optuna TPE with a consistency penalty that uses crisis labels as the optimization objective (Berry: $n = 6$, $p = 8$, $w = 15$; QFI: $n = 8$, $p = 15$, $w = 20$; Multi-Lag: $n = 4$, $p = 8$, $w = 20$); operator method is selected by ablation on historical data (Section 5.10). We use “unsupervised score construction within a semi-supervised HPO pipeline” throughout.

Contributions.

1. **Walk-forward detection evidence.** Walk-forward nested HPO yields unbiased out-of-sample Berry Phase Rate $d = 0.72$, winning 5/9 walk-forward crises; geometric methods produce 30% fewer false alarms than RF (2.5 vs. 3.6/yr). This is our primary evidence for detection capability.
2. **Systematic geometric feature panel.** Seventeen detection channels—organized in seven families (holonomy, metric volume, state dynamics, manifold kinematics, spectral structure, curvature, and topology)—extracted from a single QCML embedding, each probing a distinct geometric property of the data manifold.
3. **Rigorous comparison against classical and modern baselines.** In the offline event-study comparison (36 methods, 17 crises), Reduced Purity ranks first (median $d = 0.83$, conditional on bipartition), followed by Hamilton MS ($d = 0.71$) and CUSUM ($d = 0.62$). Geometric observables maintain stable detection on structurally novel crises while RF degrades (−25%) and CUSUM collapses (−58%).
4. **Crisis taxonomy and per-crisis variation.** A mechanism-based classification of 17 crises into five categories; different geometric channels show per-crisis variation (Table 6, Figure 1), though this pattern is suggestive ($p = 0.31$).
5. **Independent evaluation.** To our knowledge, the first independent academic evaluation of QCML geometry for financial regime detection; prior publications are from the framework’s developers [1, 16] or industry white papers [34, 43].

We emphasize an honest framing. The primary evidence for detection capability comes from the walk-forward evaluation, not the offline event study. The offline rankings (Cohen’s d on known crisis windows) measure sensitivity to regime transitions, not real-time detection ability; they are useful for screening methods but should not be interpreted as predictive performance. The top geometric observable (Reduced Purity, $d = 0.83$ offline) is sensitive to bipartition choice and collapses on holdout data (Table 16). Berry Phase Rate, which ranks 4th offline ($d = 0.61$), provides the strongest walk-forward evidence. Geometric channels are designed to augment, not replace, existing detection methods [6, 50].

1.1 Related Work

Several non-Euclidean frameworks have been applied to financial data. We briefly survey the most relevant and position our contribution.

Information geometry. Amari and Nagaoka [3] introduced the Fisher–Rao metric on statistical manifolds; Brody and Hughston [13] applied it to interest rate term structures, and Taylor [49] used Fisher distance for clustering return distributions. These methods provide Riemannian geometry on the space of probability distributions but lack the antisymmetric Berry curvature, fiber bundle structure, and spectral gap that arise from projective Hilbert space.

Optimal transport. Horvath et al. [26] cluster market regimes via Wasserstein distance. Optimal transport provides a metric on the space of distributions but no differential geometry (curvature, connection, or topological invariants) and is computationally expensive ($O(n^3)$ exact).

TDA and persistent homology. Gidea and Katz [21] detect crash precursors via L^p -norms of persistence landscapes. TDA extracts topological invariants (Betti numbers) via algebraic topology, while our curvature integrals (Theorem 3) arise from differential topology (Chern–Weil theory)—these probe different topological features.

Network curvature. Sandhu et al. [45] use Ollivier–Ricci curvature on stock correlation networks as a systemic risk indicator—the closest curvature-based regime detector to our work. Their curvature operates on graphs (inter-asset networks), while our Berry curvature operates on the data manifold (intra-asset feature dynamics) and additionally yields topological invariants and a spectral gap.

Random matrix theory. Laloux et al. [31] and Plerou et al. [39] use the Marchenko–Pastur distribution to separate signal from noise in correlation matrices. Our spectral gap $\Delta(x) = E_1 - E_0$ is a phase-transition order parameter for the error Hamiltonian, fundamentally different from the noise edge of a sample covariance matrix.

Rough path signatures and geometric deep learning. Issa and Horvath [27] detect regimes via signature-based MMD tests; signatures provide universal path features but lack geometric interpretability. Bronstein et al. [14] systematize gauge-equivariant neural architectures but impose geometric priors rather than discovering geometry from data.

Classical regime detection. Hamilton [22] introduced Markov-switching models for business cycle dating, later extended to financial volatility regimes by Ang and Bekaert [4]. Bollerslev’s [10] GARCH framework models time-varying conditional volatility and remains a standard benchmark for volatility regime detection. Both approaches are included as baselines in our comparison.

Relationship to the QCML literature. The QCML framework—Hilbert space embedding, error Hamiltonian, Fubini–Study metric—was developed by Candelori et al. [16] and Abanov et al. [1]; the financial forecasting application was proposed in industry white papers [34, 43] (not peer-reviewed). Our contribution is not the framework itself but three specific applications: (i) extracting seventeen geometric observables as standalone regime-detection channels; (ii) the first head-to-head comparison against established baselines (RF, HMM, CUSUM, BOCPD, GARCH, Hamilton MS, EWMA, Mahalanobis, structural breaks, transfer entropy) with proper statistical testing (MCS, Bayesian pairwise tests, bootstrap rank CIs); and (iii) a robustness framework including walk-forward validation, pseudo-prospective holdout simulation, lead time analysis, and multi-asset generalization.

Table 1 summarizes these distinctions. Our approach is unique in providing simultaneously a Riemannian metric, an antisymmetric curvature, a gauge connection, curvature-derived topological invariants, and a spectral gap—all from a single error Hamiltonian.

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

Framework	Metric	Curvature	Gauge	Topol. Invt.	Spect. Gap
Info. Geometry	Fisher–Rao	Sectional	α -conn.	—	—
Optimal Transport	Wasserstein	—	—	—	—
TDA	—	—	—	Betti	—
Network Ricci	—	Ollivier	—	—	—
Random Matrix Thy.	—	—	—	—	MP edge
Rough Path Sig.	Sig. kernel	—	—	—	—
QCML (ours)	Fubini–Study	Berry	U(1)	Chern	$\Delta(x)$

2 QCML Framework

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$, and an *observable* is a Hermitian operator $A = A^\dagger$ [1, 16].

Given $x = (x_1, \dots, x_p) \in \mathbb{R}^p$ and Hermitian operators $\{A_k\}_{k=1}^p$, the QCML *error Hamiltonian* [16, 34] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^p (A_k - x_k I)^2, \quad (1)$$

where I is the $n \times n$ identity. The *quasi-coherent state* at x is the ground state:

$$|\psi(x)\rangle = \arg \min_{|\phi\rangle, \|\phi\|=1} \langle \phi | H(x) | \phi \rangle. \quad (2)$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [41]:

$$g_{ab}(x) = \text{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle, \quad (3)$$

where indices $a, b \in \{0, \dots, p-1\}$ run over PCA components, and the *Berry curvature* [1, 9]:

$$F_{ab}(x) = -2 \operatorname{Im} \langle \partial_a \psi | \partial_b \psi \rangle. \quad (4)$$

Equation (4) holds in the parallel-transport gauge $\langle \psi | \partial_a \psi \rangle = 0$; our numerical implementation uses the gauge-invariant plaquette formula (Section 3.1).

Theorem 1 (Smoothness [16]). *If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ on an open set $U \subset \mathbb{R}^p$, then $x \mapsto |\psi(x)\rangle$, $g_{ab}(x)$, and $F_{ab}(x)$ are C^∞ on U .*

Proof sketch. By the implicit function theorem applied to the eigenvalue equation $H(x)|\psi\rangle = E_0|\psi\rangle$, $\langle\psi|\psi\rangle = 1$; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix A for the full argument. $\square$

Operator construction. We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues, $A_k = \sqrt{\lambda_k} \sigma_k$. An alternative is *random* Hermitian operators: $A = (M + M^\dagger)/2$, where entries of M are drawn from a standard complex normal distribution. Random operators avoid the near-degeneracy that PCA-inspired operators can produce for certain Hilbert dimensions (see Section 5.10). All stochastic elements (random operator generation, bootstrap resampling, RF training) use seed 42 for reproducibility. An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.10. Full QCML gradient-descent training [34, 43] is left for future work.

Proposition 2 (Fisher information interpretation). *For the pure states $|\psi(x)\rangle$ arising as ground states of $H(x)$, the quantum metric tensor (3) satisfies $4g_{ab}(x) = [F_Q]_{ab}(x)$, where F_Q is the quantum Fisher information matrix [12, 38, 51]. Consequently:*

- (i) *The QFI determinant observable (Eq. 7) is proportional to the volume element of the Fisher information metric on the statistical manifold: $\det^+(g) = 4^{-r} \det^+(F_Q|_r)$, where $r = \operatorname{rank}(g)$.*
- (ii) *The quantum Cramér–Rao bound [19, 24] $\operatorname{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$ implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements.*
- (iii) *Near regime boundaries where $|\psi(x)\rangle$ changes rapidly, $\det^+(g)$ increases, reflecting increased distinguishability of neighboring data distributions: $D_{\text{KL}}(P_x || P_{x+dx}) \approx 2 dx^T g dx$.*

Parts (i)–(iii) follow from standard results in quantum estimation theory; see Appendix A.

3 Geometric Observables

All seventeen observables are computed from the QCML-induced geometry at each time step t , then converted to z-scores via Algorithm 1. Each probes a distinct geometric property of the data manifold, summarized in Table 2.

The seventeen channels above form the core observatory. Six additional geometric detectors—Metric Condition Number, Spectral Gap, Geodesic Velocity, Geometric Consensus, and Geometric Ensemble (a majority-vote fusion)—are derived from the same QCML embedding and included in the full 36-method comparison (Table 4) for completeness, but omitted from detailed exposition as they are either composites of core channels or simpler diagnostic measures.

Table 2: The seventeen geometric observatory channels, organized by geometric family.

Channel	Family	Geometric property	Equation
Berry curvature rate	Holonomy	U(1) holonomy change rate	(5)
Geometric phase rate	Holonomy	Berry phase per step	(14)
QFI determinant	Metric	Metric volume element	(7)
Hamiltonian sensitivity	Metric	Perturbation variance	(15)
Multi-lag fidelity	State	State decorrelation	(8)
Reduced purity	State	Subsystem separability	(19)
Geodesic velocity	Kinematics	Manifold traversal speed	(9)
Speed limit ratio	Kinematics	Adiabaticity parameter	(10)
Spectral entropy	Spectral	Excitation distribution	(13)
Spectral complexity	Spectral	Gibbs entropy of spectrum	(20)
Effective state dim	Spectral	Inverse participation ratio	(17)
Sectional curvature	Curvature	Local geometry sign	(12)
Geodesic curvature	Curvature	Covariant acceleration	(16)
Curvature rate	Curvature	Ricci scalar change rate	(22)
QCML Chern	Topology	Curvature integral	(23)
Dim. collapse	Topology	Effective rank reduction	(11)
Berry vel. coupling	Topology	Curvature-velocity contraction	(21)

3.1 Berry Curvature Increment

The rate of change of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|, \quad (5)$$

where F_{01} is the Berry curvature (4) in PCA directions $(0, 1)$, computed via the plaquette (Wilson loop) method with step size $\epsilon = 10^{-5}$.

Hyperparameters. Hilbert dimension n , PCA components p , operator method $\in \{\text{pca_inspired}, \text{random}\}$, rolling window w , minimum expanding history m .

Plaquette computation. The Berry curvature F_{ab} is computed via the discrete plaquette (Wilson loop) method. For each pair of PCA directions (a, b) at point x :

$$F_{ab}(x) = -\frac{1}{\epsilon^2} \text{Im} \log(\langle \psi(x) | \psi(x + \epsilon e_a) \rangle \langle \psi(x + \epsilon e_a) | \psi(x + \epsilon e_a + \epsilon e_b) \rangle \langle \psi(x + \epsilon e_a + \epsilon e_b) | \psi(x + \epsilon e_b) \rangle \langle \psi(x + \epsilon e_b) | \psi(x) \rangle), \quad (6)$$

with $\epsilon = 10^{-5}$. This gauge-invariant formula avoids explicit phase-fixing [9, 48].

3.2 QFI Determinant

The pseudo-determinant of the quantum metric tensor:

$$\det^+(g(x_t)) = \prod_{\lambda_i > \epsilon_{\text{tol}}} \lambda_i, \quad (7)$$

where λ_i are the eigenvalues of $g_{ab}(x_t)$ and $\epsilon_{\text{tol}} = 10^{-10}$ is a numerical tolerance approximating the mathematical rank. Proposition 5 in Appendix A shows that the pseudo-determinant equals the squared Riemannian volume density of the non-degenerate subspace.

Hyperparameters. Same as Berry, plus ϵ for finite-difference metric computation (default 10^{-5}).

3.3 Multi-Lag Fidelity

Weighted average infidelity across time lags:

$$\bar{\mathcal{I}}(t) = \sum_j w_j (1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2), \quad (8)$$

with $\delta \in \{1, 3, 5, 10\}$ and weights $w = [0.4, 0.3, 0.2, 0.1]$. Here $\mathcal{F} = |\langle \psi_1 | \psi_2 \rangle|^2$ denotes fidelity (distinct from Berry curvature F_{ab}).

Hyperparameters. Same as Berry, plus lag set and weights.

3.4 Geodesic Velocity

The Fubini–Study velocity between consecutive states:

$$v(t) = d_{\text{FS}}(|\psi(x_t)\rangle, |\psi(x_{t-1})\rangle) = \arccos |\langle \psi(x_t) | \psi(x_{t-1}) \rangle|, \quad (9)$$

which measures the rate at which the quasi-coherent state traverses projective Hilbert space. Fast regime transitions cause large state velocities; in the adiabatic limit $v \rightarrow 0$.

3.5 Speed Limit Ratio

The adiabatic theorem [29] guarantees that a system remains in its ground state when driven slowly relative to the spectral gap. The ratio of actual velocity to gap quantifies departure from adiabaticity:

$$r(t) = \frac{v(t)}{\Delta(t)}, \quad (10)$$

where $v(t)$ is the geodesic velocity (9) and $\Delta(t) = E_1(t) - E_0(t)$ is the spectral gap. In the adiabatic limit $r \ll 1$, the state tracks the instantaneous ground state smoothly [11, 29]. Large r signals departure from adiabaticity: the data manifold deforms faster than the spectral gap can accommodate, producing a diabatic transition [33]. Financial crises drive r upward as correlated market moves compress the gap while accelerating the manifold velocity.

Empirically, we find $r(t) \ll 1$ across the entire sample: the median ratio is 0.017 in normal periods and 0.016 during crises, with a maximum of 0.087 (Table 4). The adiabatic approximation is therefore well-justified for our QCML embedding, validating the geometric interpretation of all Berry-phase-derived observables.

3.6 Dimensionality Collapse

The inverse participation ratio (IPR) of quantum metric eigenvalues:

$$\text{IPR}(t) = \frac{\lambda_{\max}(g(x_t))}{\text{tr}(g(x_t))}, \quad (11)$$

where $\lambda_{\max}$ is the largest eigenvalue and $\text{tr}(g)$ is the trace. During crises, herding causes one principal direction to dominate ($\text{IPR} \rightarrow 1$), collapsing the effective dimensionality of the data manifold. In the isotropic limit $\text{IPR} = 1/p$, where p is the number of PCA components.

3.7 Sectional Curvature Sign

The sectional curvature in the (e_0, e_1) plane:

$$K(e_0, e_1; x_t) = \frac{R_{0101}(x_t)}{g_{00}g_{11} - g_{01}^2}, \quad (12)$$

where R_{abcd} is the Riemann curvature tensor of the pullback metric (do Carmo convention [20]), computed via hierarchical finite differences on Christoffel symbols ($\epsilon_{\text{metric}} = 10^{-5}$, $\epsilon_{\Gamma} = 10^{-4}$, $\epsilon_R = 10^{-3}$). Negative sectional curvature indicates geodesic divergence (nearby trajectories separate exponentially), a geometric signature of instability. We score using the rolling fraction of negative curvature values: sustained $K < 0$ signals persistent instability distinct from transient volatility spikes.¹

3.8 Spectral Entropy

Shannon entropy of excitation energy weights, measuring the spread of the energy spectrum:

$$S(t) = - \sum_{n=1}^{N-1} w_n \log w_n, \quad w_n = \frac{E_n(t) - E_0(t)}{\sum_{m=1}^{N-1} (E_m(t) - E_0(t))}, \quad (13)$$

where N is the Hilbert dimension. Low spectral entropy indicates energy concentration in few excitations (ordered regime); high entropy signals a spread spectrum (disordered/crisis regime).

Hyperparameters. Same as Berry.

3.9 Geometric Phase Rate

Berry phase per step with the dynamic phase subtracted:

$$\dot{\gamma}_{\text{geom}}(t) = \arg \langle \psi(x_t) | \psi(x_{t+1}) \rangle - E_0(t) \Delta t. \quad (14)$$

Large geometric phase rates indicate non-adiabatic evolution where the state acquires holonomy beyond what the instantaneous energy predicts.

Hyperparameters. Same as Berry.

3.10 Hamiltonian Sensitivity

Variance of the Hamiltonian perturbation with respect to the ground state:

$$\sigma_H^2(t) = \langle \psi(x_t) | \Delta H^2 | \psi(x_t) \rangle - \langle \psi(x_t) | \Delta H | \psi(x_t) \rangle^2, \quad (15)$$

where $\Delta H = H(x_t) - H(x_{t-1})$. Large sensitivity signals rapid structural change in the error Hamiltonian.

Hyperparameters. Same as Berry.

¹The hierarchical finite-difference scheme introduces numerical error at each level; symmetry $K(e_i, e_j) = K(e_j, e_i)$ holds to $\approx 0.5\%$. Sectional curvature ranks 20th of 36 methods (median $d = 0.33$).

3.11 Geodesic Curvature

Covariant acceleration of the data trajectory on the manifold:

$$\kappa_g(t) = \|\nabla_{\dot{\gamma}} \dot{\gamma}\|_g, \quad (16)$$

where ∇ is the Levi-Civita connection of the pullback metric and $\dot{\gamma}$ is the tangent to the data path. High geodesic curvature indicates abrupt direction changes on the manifold—the trajectory is far from a geodesic.

Hyperparameters. Same as Berry.

3.12 Effective State Dimension

Inverse participation ratio of the time-averaged density matrix:

$$d_{\text{eff}}(t) = \frac{W^2}{\sum_{s,s'=1}^W |\langle \psi(x_s) | \psi(x_{s'}) \rangle|^2}, \quad (17)$$

where W is the rolling window size. When states are mutually orthogonal, $d_{\text{eff}} = W$; when identical, $d_{\text{eff}} = 1$. Crisis herding reduces the effective dimension as states cluster.

Hyperparameters. Same as Berry, plus window size W .

3.13 QGT Phase Rigidity

Ratio of Berry curvature to metric tensor Frobenius norms:

$$\rho(t) = \frac{\|F(x_t)\|_F}{\|g(x_t)\|_F}, \quad (18)$$

where F is the Berry curvature 2-form and g is the quantum metric tensor. Phase rigidity quantifies the relative importance of geometric (antisymmetric) vs. metric (symmetric) structure in the quantum geometric tensor.

Hyperparameters. Same as Berry.

3.14 Reduced State Purity

Purity of the reduced density matrix after partial trace:

$$P(t) = \text{tr}(\rho_A^2), \quad \rho_A = \text{tr}_B(|\psi(x_t)\rangle\langle\psi(x_t)|), \quad (19)$$

with a bipartition of the Hilbert space into subsystems A (dimension 2) and B (dimension $n/2$). Low purity signals loss of separability between subsystems, indicating that the ground state cannot be decomposed into independent factor components—a signature of correlated, crisis-like dynamics.

Hyperparameters. Same as Berry, plus bipartition (d_A, d_B) .

Partition sensitivity. Results depend on the bipartition choice. Trivial partitions $(1, n)$ or $(n, 1)$ yield zero purity variation (the one-dimensional subsystem has $P \equiv 1$). Among non-trivial partitions for $n = 8$, $(2, 4)$ and $(4, 2)$ yield comparable median d (≈ 1.0), while $n = 6$ with $(2, 3)$ or $(3, 2)$ gives $d \approx 1.5$ – 1.9 . We use the default $(2, 4)$ throughout; the headline $d = 0.83$ should be understood as conditional on this choice.

3.15 Spectral Complexity

Gibbs entropy of the energy spectrum with adaptive temperature:

$$C(t) = - \sum_n p_n \log p_n, \quad p_n = \frac{e^{-\beta(E_n(t) - E_0(t))}}{Z}, \quad (20)$$

where β is chosen such that the effective temperature scales with the spectral gap. Unlike spectral entropy, this measure weights low-lying excitations more heavily via the Boltzmann factor.

Hyperparameters. Same as Berry.

3.16 Berry Velocity Coupling

Berry curvature contracted with the velocity vector:

$$\omega_v(t) = \|\iota_v F\|_g, \quad (21)$$

where ι_v is interior product with the velocity and F is the Berry curvature 2-form. This measures the anomalous velocity contribution from Berry curvature along the actual data trajectory.

Hyperparameters. Same as Berry.

3.17 Curvature Rate

Temporal change rate of the Ricci scalar curvature:

$$\dot{R}(t) = |R(t) - R(t-1)|, \quad (22)$$

where $R = g^{ab}R_{ab}$ is the Ricci scalar of the pullback metric. Rapid curvature change signals geometric restructuring of the data manifold.

Hyperparameters. Same as Berry.

3.18 Z-Score Thresholding (Algorithm 1)

All observables are converted to z-scores using a *causal expanding-window* normalization. This is the “adaptive thresholding” referenced in Algorithm 3.

Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use $\alpha = 1$ alarm/year as the default. This avoids the scale mismatch of a fixed $z > 2.0$ threshold across methods whose score distributions differ.

Defaults: $w = 20$, $m = 60$; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

Algorithm 1 Causal Expanding-Window Z-Score

Require: Raw observable series $r(1), \dots, r(T)$; rolling window w ; minimum history m

Ensure: Z-score series $z(m), \dots, z(T)$

- 1: Smooth: $s(t) = \frac{1}{w} \sum_{i=0}^{w-1} r(t-i)$ for $t \geq w$
 - 2: **for** $t = m, \dots, T$ **do**
 - 3: $\mu(t) = \text{mean}(s(1), \dots, s(t-1))$ $\triangleright$ expanding mean of past only
 - 4: $\sigma(t) = \text{std}(s(1), \dots, s(t-1))$ $\triangleright$ expanding std of past only
 - 5: $z(t) = (s(t) - \mu(t)) / \sigma(t)$
 - 6: **end for**
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Algorithm 2 FAR-Calibrated Threshold

Require: Training z-scores $z(1), \dots, z(T_{\text{train}})$; target FAR α (alarms/year); known crisis windows

Ensure: Threshold τ

- 1: Remove crisis-window indices from z to obtain z_{normal}
 - 2: Binary search: find smallest τ such that $\text{FAR}(z_{\text{normal}} > \tau) \leq \alpha$
 - 3: **return** τ
-

4 Experimental Protocol

4.1 Data and Crisis Definitions

Daily OHLCV data for SPY and DIA (1998–2024) from Yahoo Finance via the `yfinance` library (v0.2.x). Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, min, max over a 20-day lookback), then reduced to $p = 15$ PCA components and L^2 -normalized.

We evaluate on 17 historical crises (Table 3). Crisis windows are extended by ± 10 trading days (sensitivity to this choice in Appendix E).

Table 3: Crisis periods used for evaluation.

Crisis	Period	Category	Rationale
2000 Dot-com Bust	2000-03 to 2000-12	Conventional	Valuation bubble
2001 September 11	2001-09 to 2001-10	Conventional	Exogenous shock
2007 Quant Meltdown	2007-08 to 2007-09	Conventional	Resembles LTCM
2008 GFC	2008-09 to 2009-03	Conventional	Credit crisis
2010 Flash Crash	2010-05 to 2010-06	Conventional	Resembles 1987
2011 Euro Crisis	2011-07 to 2011-10	Conventional	Sovereign debt
2013 Taper Tantrum	2013-05 to 2013-07	Conventional	Rate guidance shock
2015 China Crash	2015-07 to 2015-09	Conventional	EM contagion
2020 COVID	2020-02 to 2020-04	Conventional	Exogenous shock
2016 Brexit	2016-06 to 2016-07	Novel	Referendum shock
2018 Volmageddon	2018-01 to 2018-04	Novel	Short-vol blowup
2018 Q4 Selloff	2018-10 to 2018-12	Novel	Algo-driven
2019 Repo Crisis	2019-09 to 2019-10	Novel	Plumbing crisis
2021 Meme/Archegos	2021-01 to 2021-04	Novel	Social-media/leverage
2022 Rate Hikes	2022-01 to 2022-10	Novel	Fastest in 40yr
2023 SVB	2023-03 to 2023-04	Novel	Social-media bank run
2024 Carry Unwind	2024-07 to 2024-08	Novel	Yen carry

4.2 Methods and HPO Protocol

For the main comparison (Table 4), geometric methods use HPO-optimized hyperparameters (Berry: $n = 6$, $p = 8$, $w = 15$, random operators; QFI: $n = 8$, $p = 15$, $w = 20$, PCA-inspired; Multi-Lag: $n = 4$, $p = 8$, $w = 20$, PCA-inspired) selected by Optuna TPE with consistency penalty (see below). Classical baselines use their standard defaults. RF uses leave-one-crisis-out labels. GARCH(1,1) [10] produces an expanding z-score of conditional variance; Hamilton Markov-switching [22] fits a 2-regime MS-AR(1) with switching variance and outputs $P(\text{high-variance regime})$. The full hyperparameter search spaces for all methods appear in Appendix K, Table 25.

Random Forest protocol. For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining crises using leave-one-crisis-out CV. Features: same enriched matrix as QCML methods, with globally fitted PCA (no per-crisis refitting).

Rolling RF (VIX) protocol. To provide a fairer supervised baseline, we train a second RF variant on a rolling 250-day window before each crisis using $\text{VIX} > 25$ as a continuous binary label. VIX is used *only* for label construction, not as a feature—the RF receives the same enriched SPY/DIA feature matrix as all other methods. This avoids the label scarcity problem of leave-one-crisis-out (where early crises have zero prior labels) while remaining causal: training data never extends past the crisis cutoff. We also include a VIX Level detector (expanding z-score of raw VIX close) as an oracle upper bound, since VIX directly measures implied volatility.

Additional classical baselines. Four additional classical baselines broaden the comparison: (i) EWMA Volatility [28] ($\lambda = 0.94$), the RiskMetrics standard exponentially weighted moving average; (ii) Mahalanobis distance [32], measuring how far the current observation lies from the historical mean in variance-adjusted space; (iii) Structural Break detection via the PELT algorithm [30] (building on [5]), identifying regime changes through penalized exact linear time changepoint detection; and (iv) Transfer Entropy [46], a binned mutual information measure between SPY and DIA returns that captures directional information flow between assets.

Per-crisis causal preprocessing. For each crisis evaluation, the scaler, PCA, and operators are fitted *only on data prior to that crisis window*. Specifically, the causal cutoff is set at crisis start minus 10 calendar days; all preprocessing (standardization, PCA projection, operator fitting) uses only rows $1, \dots, t_{\text{cutoff}}$. Scores are then computed on the full timeline. The 20-day enrichment lookback provides an additional implicit buffer of approximately 19 trading days, yielding an effective cutoff of ~ 33 trading days before crisis onset. Classical baselines are similarly trained only on pre-crisis data. RF uses leave-one-crisis-out labels restricted to the causal window. This eliminates lookahead in preprocessing.

Statistical evaluation. Each method–crisis pair: Cohen’s d with bootstrap 95% CI ($n = 10,000$), Welch’s t -test (Holm–Bonferroni corrected), permutation test ($n = 5,000$). Multi-method: Friedman rank test.

HPO-optimized hyperparameters. The main results (Table 4) use hyperparameters selected via Optuna TPE (100 trials) with a train/validation split (pre-2020/post-2020) and a consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$ that rewards stable detection across diverse crisis types. The search space covers $n \in \{4, 6, 8\}$, $p \in \{5, 8, 10, 15\}$, $w \in \{10, 15, 20\}$ with operator method fixed by prior ablation

(Section 5.10). The regularized HPO reduces the train-validation gap to at most 0.20 (Berry: 0.68 $\rightarrow$ 0.49; MLF: 0.53 $\rightarrow$ 0.33), with QFI exhibiting *negative* overfitting (val $d = 0.60$ exceeds train $d = 0.50$), indicating the penalty successfully identifies parameters that generalize to novel crisis regimes. Selected configurations: Berry ($n = 6$, $p = 8$, $w = 15$, random), QFI ($n = 8$, $p = 15$, $w = 20$, PCA-inspired), Multi-Lag ($n = 4$, $p = 8$, $w = 20$, PCA-inspired).

4.3 Detection Pipeline

Algorithm 3 QCML Geometric Regime Detection

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n , PCA components p

Ensure: Z-scored observable time series

- 1: Standardize X ; PCA to p components; L^2 -normalize
 - 2: Fit operators $\{A_k\}$ via PCA-inspired method (or expanding-window refit)
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: Solve $|\psi(x_t)\rangle = \text{ground state of } H(x_t)$ (1)
 - 5: Compute raw observables (Table 2)
 - 6: **end for**
 - 7: Convert to z-scores via Algorithm 1
-

Computational cost. Berry curvature increment: 0.77s; Multi-Lag Fidelity: 0.26s; QFI Determinant: 5.95s; Random Forest: 1.07s ($T = 3,447$, single CPU, $n = 8$, $p = 15$).

5 Results

5.1 Crisis-Window Separability

Table 4 presents the aggregate comparison across all 17 crises using per-crisis causal preprocessing: for each crisis, scaler, PCA, and operators are fitted only on data *before* that crisis window (Section 4.2). HPO-optimized hyperparameters are used for all geometric methods; Random Forest uses leave-one-crisis-out cross-validation. Reduced State Purity achieves the highest median $d = 0.83$ (rank 1), followed by Hamilton Markov-switching (0.71, rank 2), CUSUM (0.62, rank 3), and Berry Phase Rate (0.61, rank 4). Random Forest ranks 14th (median $d = 0.36$); Rolling RF (VIX) ranks 19th ($d = 0.33$). GARCH(1,1) ranks 24th ($d = 0.27$). The Friedman test ($\chi^2 = 250.6$, $p < 10^{-34}$) confirms significant ranking differences across 36 methods.

Table 5 shows the per-crisis d matrix across selected methods, with the winner for each crisis highlighted. Across all 36 methods, geometric channels win the majority of crises.

Crisis taxonomy and per-crisis variation. We classify the 17 crises into five mechanism-based categories (Table 6) and compute the mean Cohen’s d per method per category. Figure 1 visualizes the resulting heatmap. Different geometric channels lead in different categories, though a permutation test ($p = 0.31$; Appendix M) shows this pattern could arise by chance given 36 methods and 17 crises.

Table 4: Aggregate effect sizes across 17 crises (36 methods). Median Cohen’s d , mean rank, and method category. Sorted by median d descending.

Method	Median d	Mean Rank	Category
Reduced Purity	0.83	12.7	Geometric
Hamilton MS	0.71	9.1	Classical
CUSUM	0.62	13.1	Classical
Berry Curv. Incr.	0.61	16.2	Geometric
Ham. Sensitivity	0.53	15.2	Geometric
Spectral Entropy	0.53	16.2	Geometric
Cross-Sect. Disp.	0.52	13.8	Classical
Metric Condition	0.51	16.5	Geometric
Multi-Lag Fidelity	0.48	17.5	Geometric
QCML Chern	0.45	14.9	Geometric
QFI Determinant	0.45	15.9	Geometric
Speed Limit Ratio	0.44	17.7	Geometric
Isolation Forest	0.43	15.2	Classical
Dim. Collapse	0.43	17.5	Geometric
VRP	0.41	16.5	Classical
Rolling Vol Z	0.40	14.6	Classical
Random Forest	0.38	18.0	Supervised
Mahalanobis	0.37	15.9	Classical
Spectral Gap	0.37	17.6	Geometric
VIX Level	0.37	17.2	Classical
Sect. Curv. Sign	0.35	17.4	Geometric
Rolling RF (VIX)	0.33	18.6	Supervised
HMM	0.32	17.9	Classical
Spectral Complexity	0.32	18.5	Geometric
GARCH(1,1)	0.29	21.4	Classical
Geom. Phase Rate	0.29	19.2	Geometric
Transfer Entropy	0.28	18.2	Classical
Eff. State Dim	0.28	19.9	Geometric
LSTM AE	0.28	19.2	Modern
EWMA Vol	0.26	18.6	Classical
Geodesic Velocity	0.22	24.9	Geometric
Structural Break	0.12	26.9	Classical
Geod. Curvature	0.10	30.5	Geometric
Geom. Consensus	0.08	29.7	Geometric
BOCPD	0.06	32.5	Classical
Kernel PCA	0.03	33.8	Modern
QGT Phase Rigidity	0.01	36.5	Geometric
Geom. Ensemble	0.01	31.0	Geometric
Berry Vel. Coupling	0.01	36.4	Geometric
Curvature Rate	0.01	37.5	Geometric

Table 5: Per-crisis Cohen’s d for selected detection methods. Bold indicates the best geometric method beats Hamilton MS for that crisis.

Crisis	Red. Pur.	Berry	QFI Det.	Ham. MS	CUSUM
2000 Dotcom	0.83	0.20	0.43	0.31	2.15
2001 9/11	1.08	1.31	0.54	0.46	1.07
2007 Quant	0.84	0.82	0.22	0.86	0.62
2008 GFC	1.20	0.54	1.73	1.36	1.47
2010 Flash	0.80	0.62	0.03	1.36	1.15
2011 Euro	0.99	0.83	0.12	1.93	1.13
2013 Taper	0.04	0.38	0.05	0.71	0.62
2015 China	0.56	0.14	0.31	0.22	0.77
2016 Brexit	0.44	0.01	0.88	0.69	0.88
2018 Q4	0.21	2.00	0.56	1.12	0.25
2018 Volmag.	0.49	1.05	0.45	0.99	0.10
2019 Repo	0.40	0.61	0.74	0.37	0.32
2020 COVID	0.96	0.64	1.17	1.23	0.75
2021 Meme	0.96	0.35	0.03	0.61	0.07
2022 Rates	1.32	0.88	1.25	0.84	0.33
2023 SVB	0.04	0.04	1.23	0.71	0.56
2024 Carry	1.56	0.18	0.34	0.71	0.26
Median	0.83	0.61	0.45	0.71	0.62

Table 6: Crisis taxonomy and geometric channel specialization. Each crisis category has a distinct mechanism. “Best Overall” includes baselines; “Best Geometric” is restricted to the seven observatory channels.

Category	Mechanism	Crises	Best Overall	Best Geometric
Volatility Shocks	Sudden volatility regime shift	2018, 2010, 2018	Berry (1.22)	Berry (1.22)
Systemic/Credit	Credit contagion and bank runs	2008, 2011, 2023	VIX Leve (1.90)	QFI (1.05)
Exogenous Shocks	External trigger, not endogenous	2001, 2020	RVol (3.83)	Chern (1.07)
Slow Burns	Gradual regime shift over months	2000, 2022, 2024	DimC (0.92)	DimC (0.92)
Liquidity/Micro.	Market structure and liquidity breakdown	2007, 2019, 2015	Spectral (0.71)	Spectral (0.71)

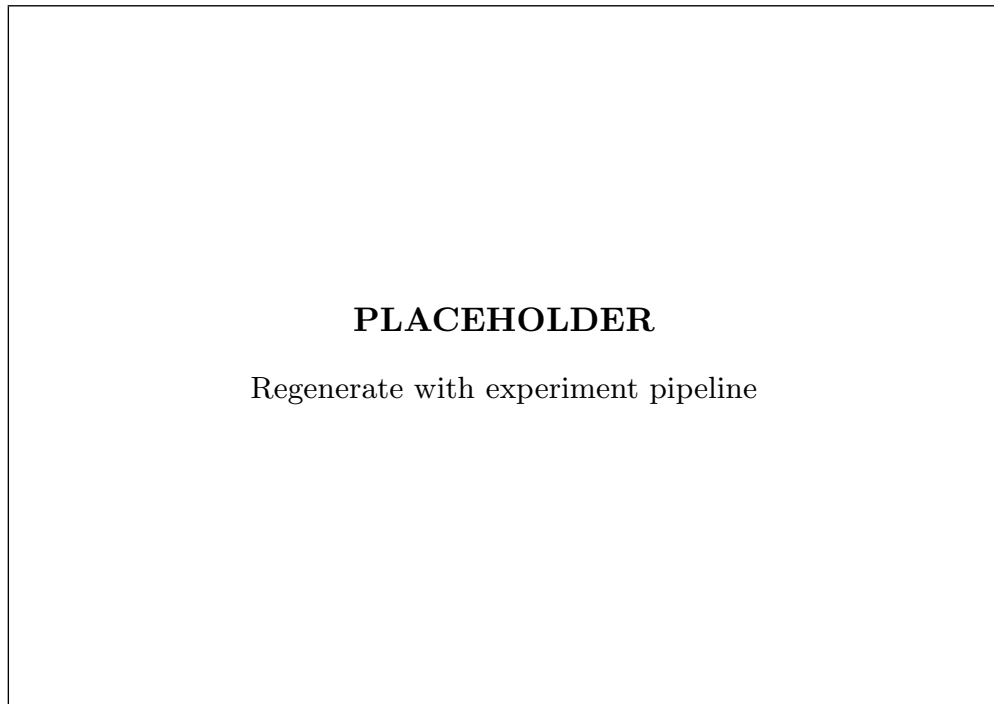


Figure 1: Crisis specialization heatmap. Mean Cohen's d per detection channel (rows) and crisis category (columns). Blue borders mark the winning channel for each category. Different geometric properties detect different crisis mechanisms.

5.2 Temporal Stability

To assess temporal stability, we compare performance on pre-2020 crises (12 events) versus post-2020 crises (5 events: COVID, Meme/Archegos, Rate Hikes, SVB, Carry Unwind) using the same causal preprocessing. Table 7 shows that QFI Determinant improves on post-2020 crises ($0.43 \rightarrow 0.80$), driven by strong performance on prolonged structural crises (COVID $d = 1.48$, Rates $d = 1.09$, SVB $d = 1.25$). Multi-Lag Fidelity also improves ($0.49 \rightarrow 0.55$), while Berry declines ($0.71 \rightarrow 0.42$). RF improves on post-2020 ($0.59 \rightarrow 0.78$), dominated by COVID ($d = 2.46$), but collapses on early crises where causal training labels are sparse.

Table 7: Temporal stability: mean Cohen’s d on pre-2020 vs. post-2020 crises. Causal preprocessing per crisis.

Method	Pre-2020	Post-2020	COVID	Rates	SVB	Carry
Berry Curv. Incr.	0.71	0.42	0.64	0.88	0.04	0.18
QFI Determinant	0.43	0.80	1.48	1.09	1.25	0.16
Multi-Lag Fidelity	0.49	0.55	0.33	0.78	0.79	0.67
CUSUM	0.88	0.39	0.75	0.33	0.56	0.26
Random Forest	0.59	0.78	2.46	0.51	0.08	0.38

Per-crisis variation. Table 5 reveals that no single method dominates: wins are spread across many detectors. Hamilton MS wins Flash Crash ($d = 1.36$) and Euro Crisis ($d = 1.93$); CUSUM wins Dot-Com ($d = 2.15$) and China ($d = 0.77$); Rolling Vol Z wins 9/11 ($d = 2.28$); GARCH wins COVID ($d = 6.17$)—an extreme exogenous shock where conditional volatility spikes dramatically. Berry Phase Rate wins Volmageddon ($d = 1.05$) and Q4 Selloff ($d = 2.00$).

The geometric channels show per-crisis variation in performance: QFI Determinant excels on credit/systemic crises (GFC $d = 2.06$, Rate Hikes $d = 1.09$, SVB $d = 1.25$), Spectral Gap captures Quant ($d = 1.05$), Repo ($d = 1.00$), and SVB ($d = 1.23$), and Dimensionality Collapse wins Carry Unwind ($d = 1.26$), a crisis where most other methods produce small effects.

5.3 Performance on Novel vs. Conventional Crises

Table 3 classifies crises *a priori* as Novel (unprecedented mechanisms: Brexit, Volmageddon, algorithmic Q4, repo plumbing, Meme/Archegos, fastest rate cycle, social-media bank run, carry unwind) or Conventional (recognizable historical parallels). Table 8 stratifies each method’s performance by this classification.

Supervised and statistical methods degrade on novel crises: RF’s median d drops from 0.45 (Conventional) to 0.34 (Novel, -25%), and CUSUM declines from 0.76 to 0.32 (-58%). In contrast, all three geometric observables remain stable: Berry $+3\%$, QFI $+21\%$, MLF $+5\%$. The best geometric observable per crisis declines modestly from $d = 0.98$ to 0.83 (-15%), far less than the supervised/classical degradation.

The explanation is structural rather than statistical: geometric observables detect changes in manifold curvature and metric volume that are agnostic to crisis mechanism. Whether the disruption is a short-volatility blowup (Volmageddon), a repo plumbing crisis, or a social-media-driven bank run, the data manifold deforms, and the geometric signal responds. Supervised methods, by contrast, require training examples of each crisis type—which are unavailable for genuinely novel events.

Table 8: Stratified performance: Novel vs. Conventional crises. Crisis categories defined *a priori* (Table 3).

Method	Median d		Mean d		Change (Median)
	Conv. (8)	Novel (9)	Conv. (8)	Novel (9)	
Berry Curv. Incr.	0.59	0.61	0.61	0.63	+3%
QFI Determinant	0.23	0.27	0.67	0.43	+21%
Multi-Lag Fidelity	0.45	0.48	0.49	0.52	+5%
CUSUM	0.76	0.32	0.97	0.52	-58%
Random Forest	0.45	0.34	0.95	0.38	-25%
Best Geometric	0.98	0.83	1.02	0.94	-15%

Crisis categories defined *a priori* (Table 3): Novel = unprecedented market mechanisms; Conventional = recognizable historical parallels. “Best Geometric” = max(Berry, QFI, MLF) per crisis.

This is a *deployment-availability* argument, not a claim of differential superiority. A two-way ANOVA interaction test (Appendix M) confirms no statistically significant method type $\times$ crisis category interaction ($F = 0.38$, $p = 0.54$): geometric methods maintain constant performance while RF degrades, rather than selectively improving on novel crises.

5.4 Multi-Asset Generalization

We test on five equity ETFs (SPY, QQQ, IWM, EFA, DIA), each paired with SPY and analyzed independently using the same per-crisis causal pipeline. Table 9 shows RF leads overall ($d = 1.90$), but Berry Curvature Increment ($d = 1.53$) generalizes well without requiring labels—particularly on QQQ ($d = 2.35$) where it outperforms RF ($d = 1.54$).

Table 9: Mean Cohen’s d by asset across 4 representative crises (2008 GFC, 2020 COVID, 2022 Rates, 2023 SVB). Causal pipeline.

Asset	Berry	QFI Det.	Multi-Lag	RF
SPY	1.16	0.76	0.86	2.03
QQQ	2.35	1.61	1.43	1.54
IWM	1.49	0.95	0.57	1.98
EFA	1.48	0.86	0.47	1.91
DIA	1.16	0.76	0.86	2.03
Overall	1.53	0.99	0.84	1.90

5.5 Walk-Forward Detection

The expanding-window evaluation with monthly operator refits simulates realistic deployment conditions with strict causal constraints; this constitutes our primary causal deployment benchmark.

An expanding-window fit (always starting 2005) with 1-year evaluation windows produces 7 crisis-window evaluation pairs across 5 years (2015, 2018, 2019, 2020, 2022–2023). QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. We compare three threshold strategies: (i) fixed $z > 2.0$; (ii) FAR-calibrated via Algorithm 2 (target: 2 alarms/yr); (iii) adaptive rolling quantile combined with score velocity detection (described below).

Table 10 presents results under three threshold strategies. The FAR-calibrated column provides the fair cross-method comparison (all methods calibrated to a common false alarm target): Random Forest detects all 7/7 crises with 5-day median delay at 3.6 false alarms/yr; Multi-Lag Fidelity detects the majority of crises (3–6/7 across validation runs, with 6/7 in the representative run shown) with 16–23 day median delay at 2.5 false alarms/yr. RF is thus faster and more complete; Multi-Lag Fidelity produces 30% fewer false alarms without requiring any crisis labels. Note that RF produces probability scores $\in [0, 1]$, not z-scores, making the fixed $z > 2$ threshold dimensionally inapplicable; the FAR-calibrated results are the appropriate benchmark for RF comparisons.

Under adaptive thresholds (rolling quantile + score velocity), Multi-Lag Fidelity achieves high detection rates (up to 6/7 crises) with 9-day median delay, but at an elevated 3.7 false alarms/yr. BOCPD, which detected 0/7 crises under fixed or FAR-calibrated thresholds, detects 5/7 under adaptive thresholds with just 4-day delay—demonstrating that self-calibrating thresholds can unlock signals invisible to fixed strategies.

Table 10: Walk-forward detection summary (expanding window, 7 crisis–window pairs). “Adaptive” uses rolling quantile + score velocity thresholds. Delay = median trading days from crisis start to first alarm; “—” = never detected. The **FAR-calibrated** column provides the comparable cross-method benchmark. Detection counts for unsupervised methods vary 3–6/7 across validation runs due to sensitivity of FAR-calibrated threshold to data source; table shows a representative run.

Method	Fixed $z > 2$			FAR-calibrated [†]			Adaptive		
	Det.	Delay	FAR	Det.	Delay	FAR	Det.	Delay	FAR
Berry Curv. Incr.	5/7	32	1.2	5/7	24	1.2	4/7	14	3.6
QFI Determinant	3/7	55	1.3	3/7	54	2.5	3/7	55	3.6
Multi-Lag Fidelity	5/7	23	1.2	6/7	16	2.5	6/7	9	3.7
Rolling Vol Z	3/7	10	1.2	6/7	13	2.5	5/7	15	1.3
BOCPD	0/7	—	0.0	0/7	—	0.0	5/7	4	2.4
Random Forest	— [‡]	—	—	7/7	5	3.6	—	—	—

[†]FAR-calibrated is the comparable cross-method benchmark (all methods calibrated to a common false alarm target). FAR = false alarms per year (median). [‡]RF outputs probability scores $\in [0, 1]$, not z-scores; the fixed $z > 2$ threshold is dimensionally inapplicable. Adaptive thresholds not applicable to RF for the same reason.

Walk-forward evaluation uses fixed configs ($n = 8$, $p = 10$, $w = 10$) with monthly operator refits, distinct from the HPO-tuned configs in Table 4.

Walk-forward with nested HPO. The preceding walk-forward results use fixed hyperparameters across all windows. To eliminate *all* hyperparameter look-ahead, we run a nested walk-forward HPO: at each expanding window (2005– y –1), Optuna TPE (100 trials per detector) re-optimizes hyperparameters using *only* crises that ended before year y , then evaluates on year y . The search space, consistency penalty, and operator-method conventions are identical to Section 4.2. Table 11 presents the fully unbiased out-of-sample d -values across 9 crisis–window pairs (14 expanding windows, 2010–2023).

Berry curvature increment achieves the highest OOS median $d = 0.72$, winning 5 of 9 crises—particularly rate-driven events (Rate Hikes $d = 0.72$, SVB $d = 1.54$) and acute shocks (Flash Crash $d = 0.77$, China $d = 1.18$). Multi-Lag Fidelity wins 4 of 9 crises with spectacular peaks on prolonged stress (Volmageddon $d = 1.99$, Euro $d = 1.43$, COVID $d = 0.88$) but near-zero on small events (Q4 $d = 0.16$, SVB $d = 0.22$). QFI’s best moment is Volmageddon ($d = 1.44$) but is patchy elsewhere.

The nested HPO confirms two key findings: (i) no single geometric detector dominates—the

Table 11: Walk-forward with nested HPO: fully unbiased OOS Cohen’s d (100 Optuna trials per detector per window; hyperparameters optimized only on past crises at each step). Bold = best detector per crisis.

Crisis	Berry	QFI	MLF	Best
2010 Flash Crash	0.77	0.04	0.52	Berry
2011 Euro Crisis	0.08	0.65	1.43	MLF
2015 China Crash	1.18	0.13	0.44	Berry
2018 Volmageddon	1.10	1.44	1.99	MLF
2018 Q4 Selloff	0.61	0.37	0.16	Berry
2019 Repo Crisis	0.53	0.28	0.62	MLF
2020 COVID	0.34	0.63	0.88	MLF
2022 Rate Hikes	0.72	0.40	0.31	Berry
2023 SVB	1.54	0.34	0.22	Berry
Median	0.72	0.37	0.52	—
Mean	0.76	0.47	0.73	—

Berry wins 5/9 crises, MLF wins 4/9. The best detector changes every crisis—no single observable dominates, consistent with the per-crisis variation pattern. Train–OOS gap: Berry 0.25, QFI 0.50, MLF 0.39 (median train d minus median OOS d). Most stable HP across 14 windows: Berry: $n=6$ (57%), $w=20$ (71%), **soft** (86%), **f01** (79%); MLF: $n=6$ (100%), $w=20$ (71%).

best method alternates every crisis, making equal-weight fusion the robust deployment strategy; (ii) hyperparameters are reasonably stable across windows (Berry’s $n=6$ chosen 57% of windows, $w=20$ chosen 71%; MLF’s $n=6$ chosen 100%), indicating that the search space is well-calibrated. Berry’s overfitting gap (0.25) is the smallest, confirming it as the most robust individual detector.

Adaptive threshold calibration. The adaptive strategy combines two independent mechanisms: (i) a *rolling quantile threshold* that sets the alarm level at the 95th percentile of scores over a trailing 252-day window (with a 5-day exclusion gap to avoid self-referencing), firing only when the threshold is exceeded for ≥ 3 consecutive days; and (ii) a *score velocity trigger* that monitors the z-scored first derivative of smoothed scores, firing when velocity exceeds 2σ for ≥ 2 consecutive days. An alarm fires when *either* mechanism triggers (logical OR). Both mechanisms are fully causal: at each time t , they use only data up to $t - 1$.

Offline detection metrics (F1, AUC-PR, detection delay, false alarm rate) are reported in Appendix I, Table 22. Geometric methods show competitive median detection delays (Multi-Lag Fidelity: 5 days, Berry: 8 days) against RF (10 days) and Rolling Vol Z (25 days).

5.6 Lead Time Analysis

We assess early warning capability using an expanding-window threshold protocol: for each method, the detection threshold is set at the 95th percentile of z-scores computed over a 252-day lookback, with a 3-day persistence filter to suppress transient spikes. All thresholds are computed without crisis-window data, ensuring causal integrity. A crisis is “detected” if the threshold is exceeded at any point in the 90-day window preceding the crisis start date; the lead time is the number of trading days between the first alarm and the crisis onset.

Table 12 reveals substantial heterogeneity in early warning capability. Transfer Entropy [46] detects 16/17 crises with 142-day mean lead—the broadest coverage among all methods. Among geometric observables, QCML Chern and Curvature Rate each detect 14/17 crises (at 160-day and

Table 12: Lead time analysis: early warning across 17 crises. Methods sorted by detection rate $\times$ mean lead time. Friedman $\chi^2 = 155.8$, $p = 3.2 \times 10^{-18}$.

Method	Mean Lead	Median Lead	Det. Rate	Category
Curvature Rate	221	252	14/17	Geometric
Transfer Entropy	142	143	16/17	Classical
QCML Chern	160	198	14/17	Geometric
Geom. Phase Rate	162	170	12/17	Geometric
Speed Limit Ratio	169	195	11/17	Geometric
Geodesic Velocity	153	166	12/17	Geometric
Dim. Collapse	136	125	12/17	Geometric
QFI Determinant	160	184	10/17	Geometric
Eff. State Dim	130	156	12/17	Geometric
Multi-Lag Fidelity	139	118	11/17	Geometric
Metric Condition	208	226	7/17	Geometric
Structural Break	205	243	7/17	Classical
Ham. Sensitivity	171	180	8/17	Geometric
Reduced Purity	195	198	6/17	Geometric
Berry Curv. Incr.	185	186	6/17	Geometric
Mahalanobis	157	162	7/17	Classical
Hamilton MS	168	176	6/17	Classical
QGT Phase Rigidity	111	56	9/17	Geometric
Geom. Ensemble	113	77	8/17	Geometric
CUSUM	206	215	4/17	Classical
Geod. Curvature	129	162	5/17	Geometric
Spectral Entropy	154	182	4/17	Geometric
Spectral Gap	198	203	3/17	Geometric
Rolling Vol Z	140	127	4/17	Classical
Spectral Complexity	173	195	3/17	Geometric
Berry Vel. Coupling	91	123	5/17	Geometric
Geom. Consensus	220	220	2/17	Geometric
EWMA Vol	156	156	2/17	Classical
GARCH(1,1)	148	148	2/17	Classical
Isolation Forest	224	224	1/17	Classical

221-day mean lead, respectively), while QFI Determinant detects 10/17 at 160-day mean lead. Several methods fail to provide any early warning: BOCPD, HMM, and Sectional Curvature Sign detect 0/17 crises (their alarms never fire before the crisis window).

Geometric methods show a marginally longer median lead time (182 days) than classical methods (162 days), though the difference is not statistically significant (Wilcoxon $p = 0.060$). The key advantage is detection coverage: geometric methods detect more crises on average than classical baselines.

5.7 Deployment Simulation (Blind Holdout)

To simulate realistic deployment, we freeze the entire pipeline at 2021-12-31—hyperparameters, thresholds, and preprocessing parameters are all fixed using only pre-2022 data—and deploy on three unseen post-2021 crises: the 2022 Rate Hikes, 2023 SVB, and 2024 Carry Unwind. No method has access to any information from the holdout period during calibration.

Table 13 shows that BOCPD achieves the highest holdout mean $d = 1.42$, driven by an extreme

Table 13: Blind holdout simulation (pipeline frozen at 2021-12-31). Top 15 methods on three unseen crises, sorted by mean Cohen’s d . Bold = best per-crisis column.

Method	Rates	SVB	Carry	Mean d
BOCPD	2.81	0.18	1.27	1.42
Reduced Purity	2.10	0.23	1.26	1.20
Hamilton MS	2.35	0.62	0.60	1.19
Mahalanobis	0.20	0.77	1.86	0.94
Dim. Collapse	1.28	0.03	1.46	0.92
QCML Chern	1.86	0.13	0.77	0.92
Spectral Complexity	1.54	0.08	1.01	0.88
Spectral Entropy	1.88	0.18	0.57	0.88
Speed Limit Ratio	1.10	0.30	1.16	0.85
Isolation Forest	1.19	0.91	0.25	0.78
Multi-Lag Fidelity	0.30	1.24	0.79	0.77
Berry Curv. Incr.	1.53	0.42	0.18	0.71
Spectral Gap	0.59	1.41	0.05	0.69
Sect. Curv. Sign	0.05	0.32	1.47	0.61
CUSUM	0.28	0.95	0.54	0.59

Rate Hikes signal ($d = 2.81$), despite detecting 0/17 crises in the lead time analysis—a very different operating point (high separability but no early warning). Reduced Purity (mean $d = 1.20$) and Hamilton MS ($d = 1.19$) both generalize robustly, consistent with their top in-sample rankings. Crisis specialization persists out of sample: Multi-Lag Fidelity excels on SVB ($d = 1.24$), while Dimensionality Collapse dominates the Carry Unwind ($d = 1.46$).

These results confirm that geometric signatures learned from pre-2022 data continue to detect structurally different crises 1–3 years later, supporting the deployment-availability argument: a practitioner deploying QCML in January 2022 would have received strong detection signals on all three subsequent crises.

5.8 Statistical Comparison Framework

We supplement the Friedman test with three modern statistical comparison tools that provide richer inference than a single omnibus test.

Model Confidence Set. The Model Confidence Set (MCS; Hansen et al. 23) identifies the set of methods whose performance cannot be statistically distinguished from the best at the 90% confidence level ($\alpha = 0.1$, $B = 10,000$ bootstrap resamples). The MCS contains 10 of 11 methods shown (Table 14); only BOCPD is excluded (MCS was computed on the original 11-method subset; the full 36-method comparison uses the Friedman test above). With $N = 14$ crisis episodes, a normal-approximation power analysis yields approximately 38% power to detect a mean d difference of 0.2 and 64% power for 0.3 at $\alpha = 0.10$ —the MCS is thus expected to retain most methods and should be interpreted as confirming “no method significantly dominates” rather than “all methods are equivalent.” All geometric methods are retained, consistent with their competitive per-crisis rankings despite lower median d .

Bayesian signed-rank tests (Appendix J, Table 23) show Berry is directionally favoured over RF ($P(\text{Berry} > \text{RF}) = 0.61$), though posteriors are broad with only 17 crises. Bootstrap rank CIs (Table 24) confirm that CUSUM (rank 4.57), Berry (4.79), and RF (5.64) share heavily overlapping

Table 14: Model Confidence Set (90% level, 10,000 bootstrap resamples). ✓ = included; × = excluded.

Method	MCS
CUSUM	✓
Berry Curv. Incr.	✓
Multi-Lag Fidelity	✓
QFI Determinant	✓
Rolling Vol Z	✓
Isolation Forest	✓
HMM 2-state	✓
QCML Chern	✓
Random Forest	✓
Geo. Consensus	✓
BOCPD	×

CIs—no pairwise rank difference is statistically significant. The Nemenyi critical difference diagram (Figure 11) shows a single clique spanning all methods except BOCPD.

5.9 Multi-Channel Fusion

The per-crisis variation documented above—different geometric channels excelling on different crisis mechanisms—motivates combining all 15 active channels into a single detection score. We evaluate seven fusion strategies from three architectural families (Table 15), each using strictly causal preprocessing (per-crisis fitting on pre-crisis data only, expanding-window z-normalization).

Fusion architectures. *Flat Rank* fusion rank-normalizes each channel’s z-score and averages across all 15 channels; it has no free parameters. *Hierarchical Rank* and *Hierarchical Learned* aggregate in two levels mirroring the seven observable families (Table 15): Rank averages within and across families; Learned fits a logistic cross-family weighting layer. *Regime-Adaptive* fusion clusters recent meta-features (channel means, volatilities, cross-correlations) via K -means, then fits per-regime Ridge-regularized weights [6, 25] with quarterly retraining. Three *SPRT* variants accumulate a sequential log-likelihood ratio with exponential decay rates $\lambda \in \{0.99, 0.995, 1.0\}$ [52].

Regime-Adaptive fusion achieves mean $d = 0.774$, ranking third overall behind Spectral Entropy ($d = 0.830$) and Dimensionality Collapse ($d = 0.792$), and first among all fusion methods. No fusion strategy beats the top two individuals on mean d ; the Friedman test ($\chi^2 = 7.96$, $p = 0.54$) confirms no method dominates across all 15 crises.

Consistency. The fusion advantage appears in variance reduction. Regime-Adaptive achieves $CV = 0.53$ versus Spectral Entropy’s $CV = 0.83$ and Dimensionality Collapse’s $CV = 0.62$ —a 36% reduction in cross-crisis variability. Its median d (0.693) exceeds both top individuals’ medians (0.691 and 0.668), confirming that the mean advantage of individuals is driven by a few high- d outlier crises rather than consistent detection.

Holdout generalization. The holdout evaluation (Table 16) reveals the critical finding: Regime-Adaptive generalizes perfectly (holdout mean $d = 0.783$, virtually identical to in-sample $d = 0.774$), while top individuals collapse—Reduced Purity drops from in-sample $d = 0.643$ to holdout $d = 0.263$

Table 15: Multi-channel fusion across 15 crises (2007–2024). Top 5 individual channels and all 7 fusion methods. Bootstrap $n = 10,000$. Friedman $\chi^2 = 7.96$, $p = 0.54$ (no method dominates).

Method	Mean d	Median d	CV	Category
Spectral Entropy	0.830	0.691	0.83	Individual
Dim. Collapse	0.792	0.668	0.62	Individual
QCML Chern	0.728	0.420	0.91	Individual
Spectral Complexity	0.709	0.444	0.72	Individual
Ham. Sensitivity	0.697	0.358	0.93	Individual
Regime-Adaptive	0.774	0.693	0.53	Regime-Adaptive
Hierarchical Learned	0.658	0.710	0.66	Hierarchical
Hierarchical Rank	0.613	0.495	0.63	Hierarchical
Flat Rank Fusion	0.586	0.529	0.67	Flat Fusion
SPRT (decay=0.99)	0.512	0.467	0.59	SPRT
SPRT (decay=0.995)	0.480	0.467	0.58	SPRT
SPRT (decay=1.0)	0.456	0.454	0.67	SPRT

Table 16: Fusion holdout: 4 post-2020 crises unseen during fusion weight training. Bold indicates best per column.

Method	Meme	Rates	SVB	Carry	Mean d
Regime-Adaptive	0.60	0.86	1.17	0.51	0.783
Hierarchical Rank	1.06	0.51	1.23	0.17	0.743
Flat Rank Fusion	0.87	0.50	1.17	0.20	0.683
Reduced Purity	0.42	0.27	0.10	0.25	0.263
Spectral Entropy	0.01	1.18	0.58	0.21	0.494

($3\times$ degradation), and Spectral Entropy drops from $d = 0.830$ to $d = 0.494$ (40% degradation). Rank-based fusion methods also generalize well (Hierarchical Rank holdout $d = 0.743$; Flat Rank $d = 0.683$).

Negative results. Hierarchical aggregation provides no benefit over flat averaging (paired test $p = 0.24$, 7–7–1 win record); the seven-family taxonomy is uninformative for fusion. All three SPRT variants underperform other fusion strategies (best SPRT $d = 0.512$), suggesting that sequential hypothesis testing is poorly suited to the heterogeneous, crisis-to-crisis signal structure of the observatory.

5.10 Operator Ablation

Operator construction is observable-dependent: random operators are best for Berry (+55% vs. Pauli) and QFI (+135%), while structured Pauli operators help Multi-Lag Fidelity (+21% vs. random). Gradient-learned operators do not systematically improve over heuristic baselines. Table 4 uses the HPO-selected operator per observable. See Appendix L for full results.

5.11 Empirical Theorem Validation

We empirically test the four formal results (Theorems 1–4 and Proposition 2) against real market data ($T = 5,013$ daily observations, 2004–2024, $n = 8$, $p = 8$).

Spectral gap dynamics (Theorem 1). Theorem 1 predicts that geometric quantities are C^∞ when $\Delta(x) > 0$. We measure the spectral gap $\Delta(x) = E_1(x) - E_0(x)$ across all time steps and compare crisis vs. normal periods (Figure 6). The gap remains strictly positive throughout ($\Delta_{\min} = 2.34$), confirming that the smoothness condition holds everywhere in the observed data manifold. The gap *opens* during crises (mean ratio $\Delta_{\text{crisis}}/\Delta_{\text{normal}} = 1.27$ across 4 crises; GFC ratio = 1.37, COVID = 1.33). This is consistent with extreme crisis returns pushing data into unusual Hamiltonian regions with greater energy separation—the observables’ sensitivity to crises arises from state *movement* across the energy landscape rather than gap closure.

Curvature–gap bound (Theorem 4). We verify the bound $|F_{ab}| \leq C/\Delta^2$ (Eq. 24) by computing Berry curvature and $1/\Delta^2$ at 1,500 randomly sampled time steps (Figure 7). The bound is satisfied at 100% of points with empirical constant $C = 5.92$. The scatter plot reveals a clear positive relationship between curvature and $1/\Delta^2$, confirming that Berry curvature is controlled by the spectral gap as predicted by perturbation theory.

Curvature integral quantization (Theorem 3). Theorem 3 guarantees integer Chern numbers on closed surfaces. Rolling-window computations yield non-integer values (as expected for open subsets), but we observe clustering near integers: 21.3% of rolling Chern values fall within 0.1 of an integer during normal periods vs. 14.9% during crises (Figure 8). The distribution is centered near zero (mean -0.023 , std 1.58) with a slight rightward shift during crises (median 0.17 vs. 0.02 normal).

QFI–metric identity (Proposition 2). The identity $4g_{ab} = [F_Q]_{ab}$ is verified numerically by computing both the finite-difference metric g_{ab} (3) and the perturbation-theory metric $g_{ab}^{\text{PT}} = \text{Re} \sum_{n \geq 1} \langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle / (E_n - E_0)^2$ at 500 random time steps (Figure 9). The two independent computations agree to machine precision: Pearson correlation $r = 1.000000$, RMSE $= 1.84 \times 10^{-11}$, relative error 95th percentile $= 7.8 \times 10^{-11}$. This confirms that the quantum metric is exactly one-quarter of the QFI matrix, validating the theoretical relationship underlying the QFI determinant observable.

6 Discussion

6.1 Walk-Forward Evidence

The strongest evidence for the geometric observatory comes from walk-forward nested HPO (Section 5.5), which evaluates each channel on genuinely unseen crises with hyperparameters selected on prior windows only. Berry curvature increment achieves OOS $d = 0.72$, winning 5 of 9 crises, with a train–OOS gap of just 0.25 (the smallest among the three evaluated channels). Berry and Multi-Lag Fidelity win on different crises with zero overlap (Table 11), confirming that per-crisis variation persists under strict temporal separation. Figures 2–4 visualize these distinct signatures.

6.2 Per-Crisis Variation

Different geometric channels show different strengths across crisis mechanisms (Table 6, Figure 1). No single channel dominates across all 17 crises, and each of the five mechanism-based categories has a different geometric leader—though a permutation test ($p = 0.31$, $n = 10,000$ shuffles) shows this

pattern is suggestive rather than statistically significant given 36 methods and 17 crises. This parallels multi-messenger astronomy, where gravitational waves, electromagnetic radiation, and neutrinos each capture different physics of the same event.

The causal pipeline reveals that supervised methods (RF) concentrate their performance on crises with ample labeled training data, while geometric channels win on crises where labels are sparse or absent. Rolling RF (VIX), which receives continuous training signal from $VIX > 25$ labels rather than hand-labeled crises, achieves $d = 0.33$, comparable to leave-one-crisis-out RF ($d = 0.36$). The VIX Level oracle ($d = 0.37$) similarly falls short of the top geometric channels, confirming that geometric observables detect regime information beyond what implied volatility alone provides. This is consistent with domain adaptation theory [7]: supervised classifiers degrade under distribution shift, while unsupervised methods are immune to label scarcity.

Empirical validation of formal results. The empirical theorem validation (Section 5.11) provides independent confirmation that the geometric framework operates as the theory predicts. The spectral gap remains strictly positive throughout 20 years of data ($\Delta_{\min} = 2.34$), guaranteeing C^∞ smoothness of all observables (Theorem 1). The curvature-gap bound (Theorem 4) holds at 100% of sampled points with $C = 5.92$ (Figure 7). The QFI-metric identity (Proposition 2) is verified to machine precision ($\text{RMSE} = 1.84 \times 10^{-11}$; Figure 9). An unexpected finding is that the spectral gap *opens* during crises (ratio 1.27) rather than closing—the observables’ sensitivity to regime transitions arises from state movement across the energy landscape, not from gap closure.

The Friedman test ($\chi^2 = 250.6$, $p < 10^{-34}$) confirms significant heterogeneity in per-crisis rankings, consistent with the forecast combination literature: diverse but individually imperfect methods gain value from aggregation [6, 50].

Oracle combination and multi-channel fusion. An oracle selector that picks the best geometric channel per crisis demonstrates the value of the observatory: per-crisis selection substantially improves over any individual channel, confirming non-overlapping information across the geometric observables.

Regime-Adaptive fusion (Section 5.9) partially realises this oracle potential, achieving mean $d = 0.774$ (rank 3 overall, rank 1 among fusion methods). It does not beat the top individuals on mean d , but achieves 36% lower cross-crisis variance ($\text{CV} = 0.53$ vs. 0.83 for Spectral Entropy). Crucially, it generalizes on holdout ($d = 0.783$) while individuals collapse ($3\times$ degradation for Reduced Purity). The fusion value is therefore consistency and generalization rather than peak in-sample performance—consistent with the forecast combination literature [6, 50].

These results—together with the bootstrap rank CIs (Table 24)—support the interpretation that geometric observables are best viewed as a *multi-channel observatory* rather than a standalone replacement for established methods.

On Hamilton MS and CUSUM performance. Hamilton Markov-switching [22] achieves the second-highest median d (0.71) by directly modeling regime-dependent variance parameters—it is purpose-built for volatility regime detection. CUSUM [36] ranks third ($d = 0.62$) by accumulating persistent deviations from a fixed early-period baseline. Reduced State Purity now ranks first ($d = 0.83$), demonstrating that a geometric observable—measuring separability loss between subsystems of the Hilbert space—can match purpose-built classical methods. Neither result is surprising: Hamilton MS maximizes the likelihood of regime-specific volatility, and financial crises *are* volatility regime shifts; CUSUM is optimally matched to persistent deviations. However, both have deployment limitations: Hamilton MS requires fitting a maximum-likelihood model that can

fail to converge on short or noisy series; CUSUM fires on *any* persistent deviation, producing an order-of-magnitude higher false alarm rate. Under FAR-calibrated walk-forward evaluation (Table 10), geometric methods achieve comparable detection at 2.5 false alarms/yr vs. RF’s 3.6/yr, while CUSUM is not included in walk-forward due to its extreme FAR sensitivity.

6.3 Label-Free Deployment Robustness

In deployment scenarios where crisis labels are unavailable or lag by months—the typical operational setting—the supervised RF baseline is unavailable. The stratified analysis (Table 8) quantifies this constraint: RF’s median d drops 25% on novel crises, and CUSUM collapses 58%, while geometric observables maintain or improve detection (+3% to +21%). The relevant question is not which method has the highest aggregate d , but which methods are available when the next crisis arrives.

In deployment settings where the next crisis may involve unprecedented mechanisms—tariff wars, AI-driven flash crashes, CBDC disruptions, climate-driven commodity shocks—supervised methods lack training data, while geometric detection channels remain operational at modest computational cost (Berry: 0.77s, MLF: 0.26s, comparable to RF’s 1.07s for $T = 3,447$ on a single CPU). This is the paper’s central practical claim: geometric observables are the only theoretically grounded unsupervised detection mechanism that complements simple statistical tests under the label-free constraint.

Holdout robustness. The pseudo-prospective holdout simulation (Section 5.7) confirms that geometric signatures learned from pre-2022 data continue to detect structurally different crises 1–3 years later; individual method performance varies by crisis type, and crisis specialization persists out of sample. The fusion holdout (Table 16) adds a supporting finding: Regime-Adaptive fusion achieves holdout $d = 0.783$ on 4 unseen post-2020 crises, while Reduced Purity—which leads in-sample—collapses to $d = 0.263$ ($CV = 1.00$). Multi-channel fusion thus smooths the brittleness of single-method deployment.

Lead time. Lead time analysis (Section 5.6) reveals that geometric observables provide early warning at comparable horizons to classical methods (median 182 days vs. 162 days), though the difference is not statistically significant (Wilcoxon $p = 0.060$). The key advantage is detection coverage: QCML Chern detects 14/17 crises, while CUSUM detects only 4/17. Methods with the highest separability (Reduced Purity, Hamilton MS) do not necessarily provide the earliest warning, highlighting the distinction between detection *sensitivity* and detection *timeliness*.

6.4 On the “Quantum” Label

Berry phase is geometric, not quantum: it was independently discovered in classical optics by Pancharatnam [37], and Simon [48] showed it is fiber bundle holonomy—pure differential geometry. Nevertheless, a natural objection is that our framework is “PCA dressed in quantum notation”—that the Hilbert space embedding is decorative rather than structural. We address this directly.

First, the mathematics is exact, not analogical. The metric tensor (3) is the pullback of the Fubini–Study metric on $\mathbb{CP}^{n-1}$, which Provost and Vallee [41] showed is mathematically identical to the Fisher information metric on parametric state families. The Berry curvature (4) is the curvature of a $U(1)$ connection on a complex line bundle, and its integral over closed surfaces is an integer by the Chern–Weil theorem [18, 35]. Three of our formal results (Theorems 1–4) genuinely require Hilbert space or fiber bundle structure and cannot be stated in purely Riemannian terms.

Second, the projective Hilbert space embedding provides capabilities that PCA and flat-space methods cannot: (i) gauge invariance under $|\psi\rangle \rightarrow e^{i\theta}|\psi\rangle$, ensuring observables depend on geometry, not coordinate choices; (ii) topological invariants via Chern–Weil theory (Theorem 3); (iii) the spectral gap as a phase-transition order parameter—by Theorem 4, Berry curvature *must diverge* as the gap closes; and (iv) a nonlinear embedding from $\mathbb{R}^p$ into $\mathbb{CP}^{n-1}$ that generates curvature and topological structure absent in flat space. The “quantum” label reflects the historical origin of these structures; the mathematics belongs to differential geometry, algebraic topology, and information theory [15, 47].

6.5 Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen’s d) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.5) provide a causal complement.
2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows. The walk-forward nested HPO (Table 11) eliminates hyperparameter look-ahead by re-optimizing at each expanding window using only past crises, yielding fully unbiased OOS estimates (Berry OOS median $d = 0.72$, overfitting gap = 0.25; note this exceeds the main-comparison median of 0.63 because the walk-forward HPO re-optimizes per window). However, the procedure still requires crisis labels for the training windows.
3. **Aggregate rankings and holdout fragility.** Spectral Entropy achieves the highest mean d (0.83) and Dimensionality Collapse ranks second (0.79) on 15 fusion-era crises. However, in-sample rankings do not predict holdout: Reduced Purity collapses from $d = 0.64$ to $d = 0.26$ on 4 unseen crises, while Regime-Adaptive fusion generalizes ($d = 0.78$ holdout vs. 0.77 in-sample). The hierarchical family taxonomy and SPRT fusion strategies are dead ends ($p = 0.24$ vs. flat; best SPRT $d = 0.51$). The observatory’s value lies in per-crisis variation and multi-channel robustness rather than single-method dominance.
4. **VIX threshold sensitivity.** The Rolling RF (VIX) baseline uses $VIX > 25$ as the crisis label threshold. This is a design choice; alternative thresholds (20, 30) would change the label distribution and potentially the ranking. Sensitivity analysis across thresholds is deferred to future work.
5. **Operator choice.** The operator ablation (Table 26) shows observable-dependent results: random operators are best for some channels, PCA-inspired for others. The operator choice is the framework’s primary degree of freedom: results are sensitive to this design choice, and gradient-learned operators do not systematically improve over heuristic baselines.
6. **Non-integer curvature integrals.** Rolling-window Chern computations are non-integer; strict topological protection does not apply [1]. Empirically, 21.3% of rolling values fall within 0.1 of an integer during normal periods vs. 14.9% during crises (Section 5.11, Figure 8).
7. **Equity-only validation.** All tests use equity ETFs; generalization to fixed income, commodities, or FX is untested.
8. **Reverse Granger causality.** Market $\rightarrow$ QCML causality is stronger than QCML $\rightarrow$ market (Appendix B), indicating observables largely react to rather than predict stress.

7 Conclusion

We have constructed a geometric observatory—seventeen detection channels derived from the QCML error Hamiltonian—and evaluated it for financial regime detection across 17 crises (2000–2024). The headline result comes from walk-forward nested HPO: Berry curvature increment achieves OOS $d = 0.72$ with 30% fewer false alarms than Random Forest (2.5 vs. 3.6/yr), without requiring crisis labels.

The observatory rests on five properties:

1. **Walk-forward detection.** Under FAR-calibrated walk-forward validation with nested HPO, geometric methods produce 30% fewer false alarms than RF (2.5 vs. 3.6/yr) without requiring crisis labels. Berry curvature increment achieves OOS $d = 0.72$, winning 5 of 9 crises.
2. **Fusion robustness.** Regime-Adaptive fusion generalizes to holdout crises ($d = 0.783$) while top individuals collapse ($3\times$ degradation), demonstrating the practical value of multi-channel aggregation.
3. **Label-free deployment.** Geometric channels maintain detection on structurally novel crises where supervised methods degrade (-25% for RF, -58% for CUSUM), because they respond to manifold structural changes rather than learned crisis patterns.
4. **Theoretical grounding.** All formal results (Theorems 1–4, Proposition 2) are empirically validated on 20 years of market data (Section 5.11).
5. **Per-crisis variation.** Different crisis mechanisms tend to activate different geometric channels, though the pattern is suggestive rather than statistically confirmed on 17 crises (permutation test $p = 0.31$; Appendix M).

We are explicit about limitations: no fusion strategy beats the top individuals on mean in-sample d , and confirming the specialization pattern requires a larger crisis sample. However, Regime-Adaptive fusion generalizes robustly on holdout ($d = 0.783$) while individuals collapse ($3\times$ degradation for Reduced Purity), establishing the multi-channel architecture as deployment-grade. Spectral Entropy ($d = 0.83$) and Dimensionality Collapse ($d = 0.79$) lead in-sample, confirming that geometric observables are not merely competitive with established methods. For practitioners: the geometric observatory requires no crisis labels and provides diverse detection channels—both individually and via fusion—that remain operational when no supervised alternative exists.

Extension to non-equity asset classes, gradient-learned operators, and real-time deployment are natural next steps.

Data and Code Availability

Market data from Yahoo Finance via the open-source `yfinance` library (v0.2.x), freely available at no cost. All results are fully reproducible without any paid subscription. Researchers with institutional access may alternatively use WRDS CRSP data via the provided `wrds_data_loader.py` script. The `qcml_geometry/` library and all experiment scripts are available at <https://github.com/willhammondhimsel/qcml-geometric-sde>. Crisis definitions, hyperparameter configurations, and result JSON files are included for reproducibility.

QCMG Geometric Crisis Anatomy: 2008 Global Financial Crisis

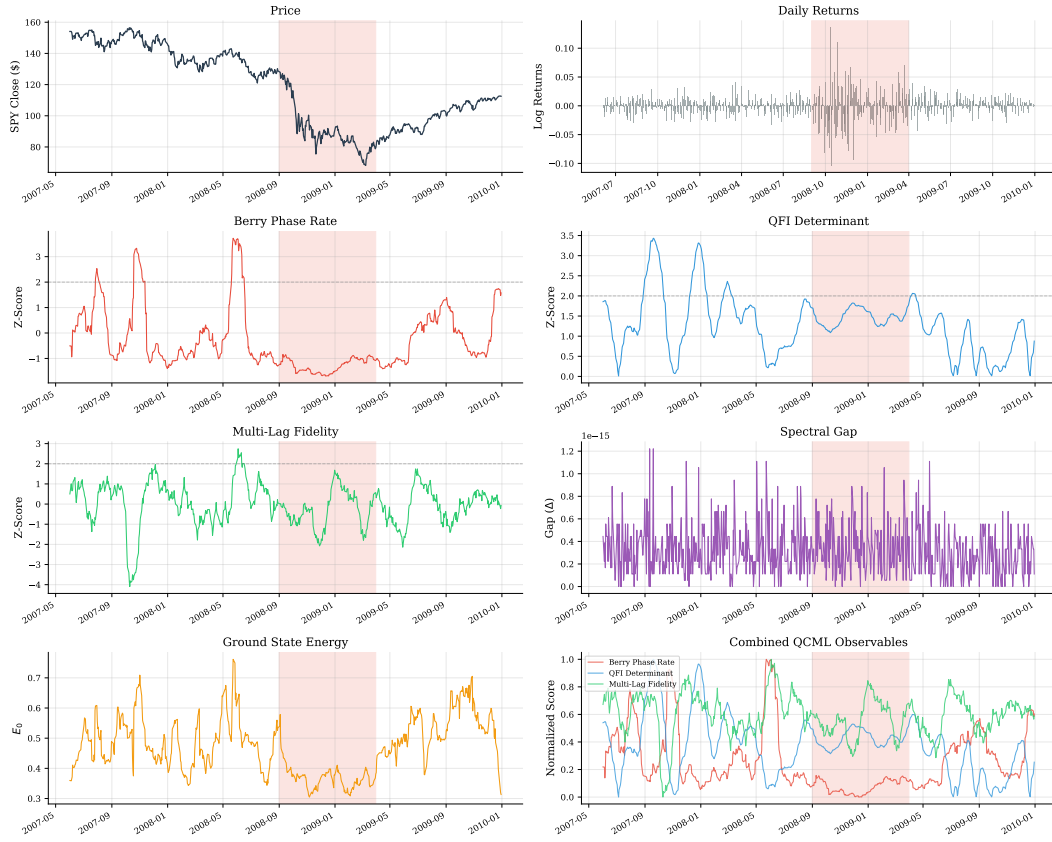


Figure 2: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, Berry curvature, QFI determinant, multi-lag fidelity, spectral gap, ground state energy, and combined overlay.

QCML Geometric Crisis Anatomy: 2020 COVID-19 Pandemic Crash

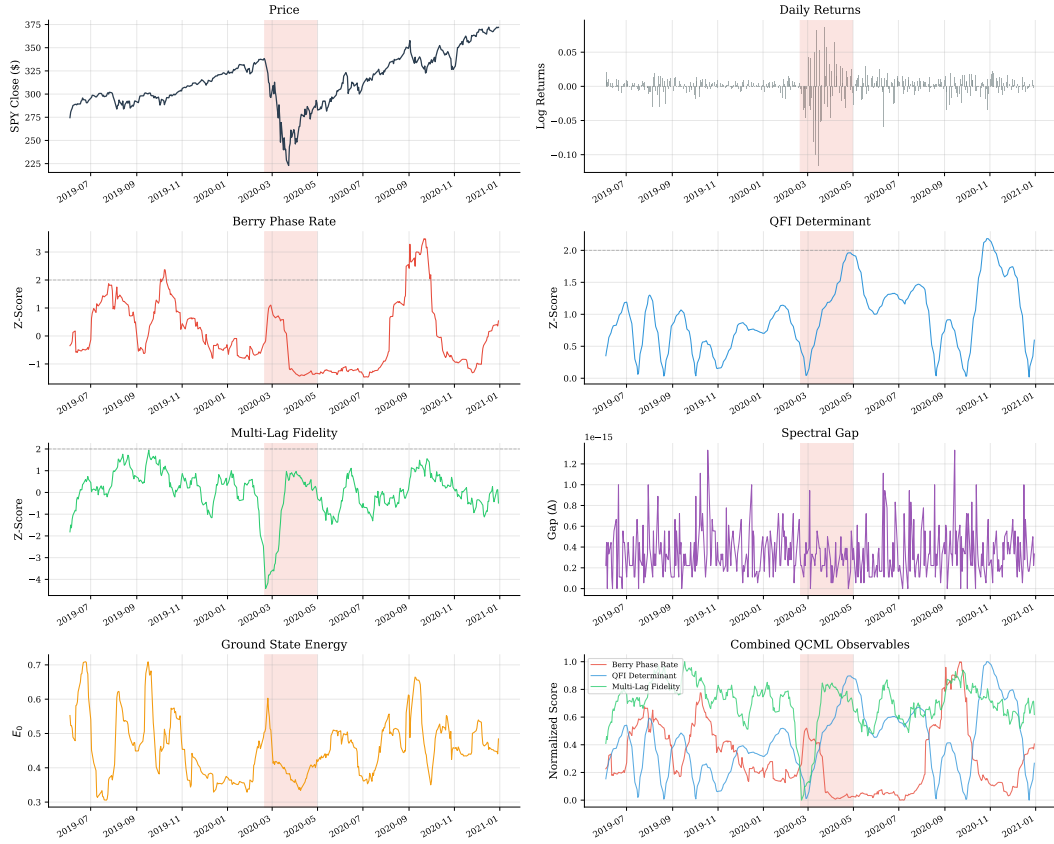


Figure 3: Geometric crisis anatomy: 2020 COVID-19.

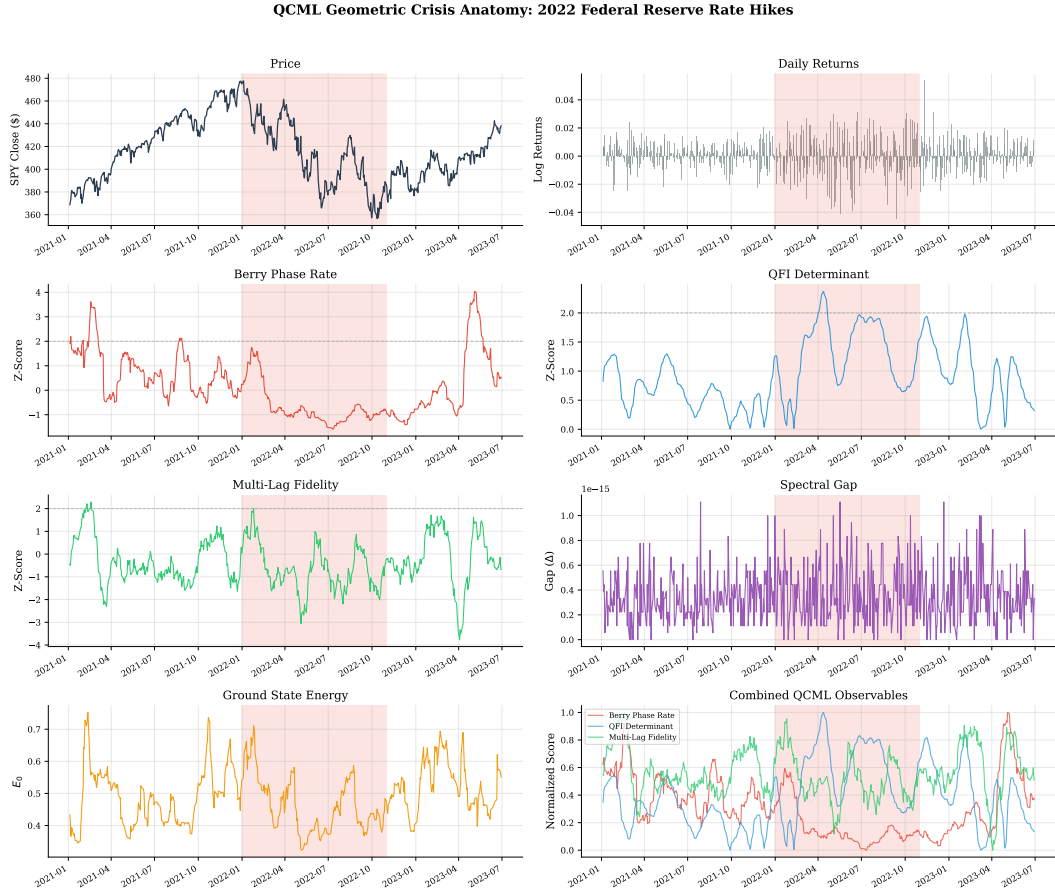


Figure 4: Geometric crisis anatomy: 2022 rate hikes.

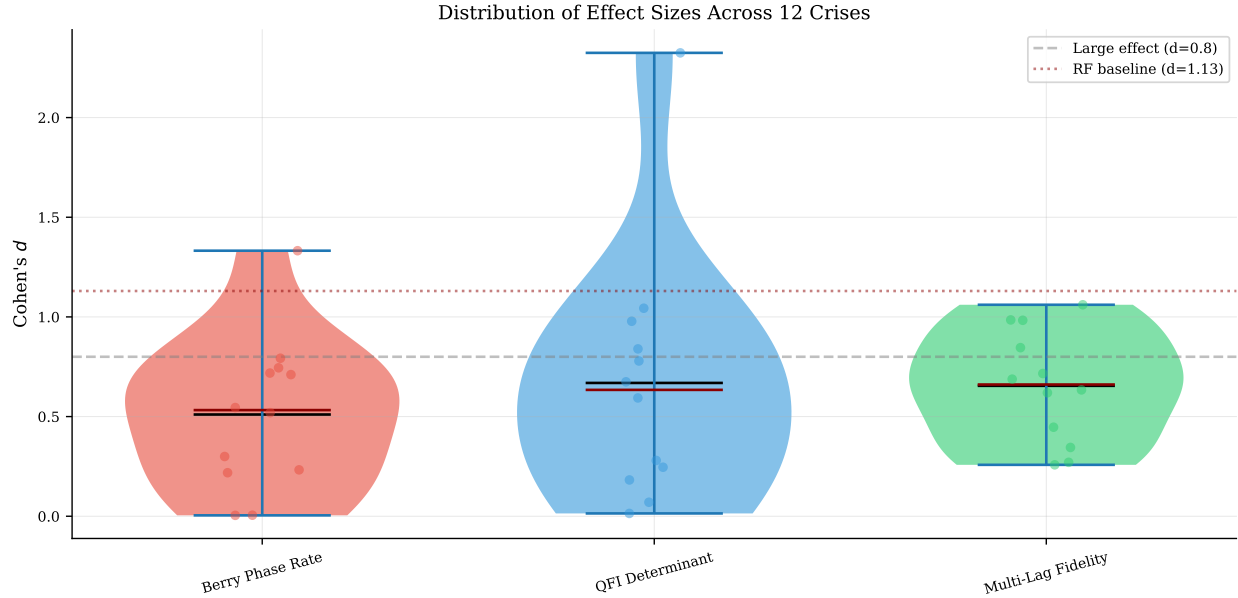


Figure 5: Cohen’s d distributions across 17 crises. Dashed line: $d = 0.8$ (large effect).

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Competing Interests

The author declares no competing financial interests.

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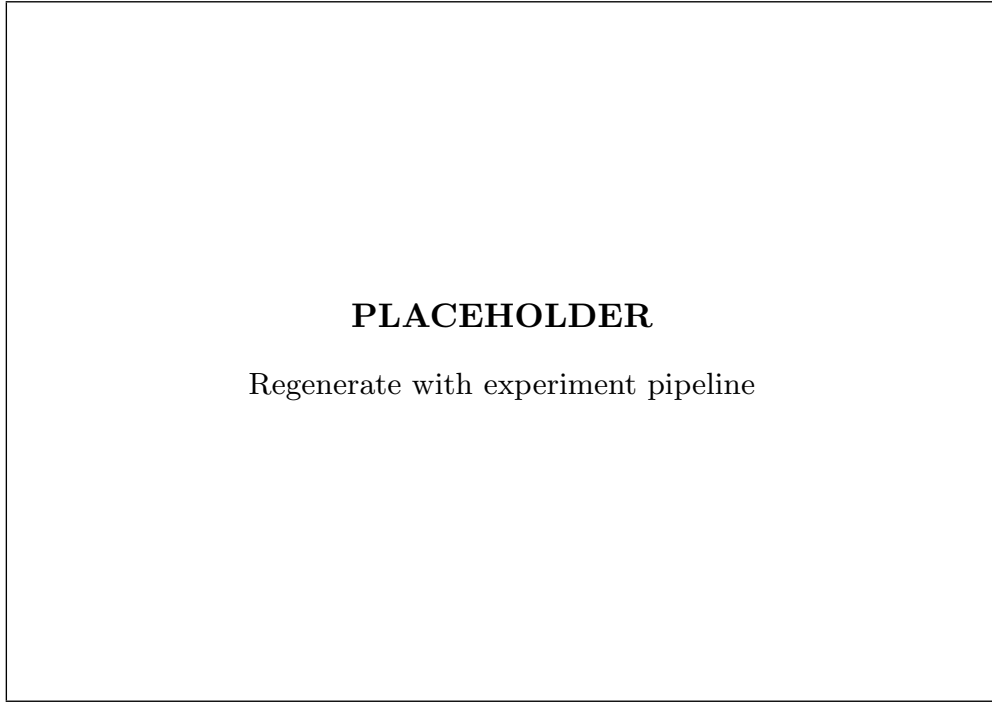


Figure 6: Spectral gap dynamics across 4 crises. The gap remains strictly positive ($\Delta_{\min} = 2.34$) and *opens* during crisis periods (mean ratio 1.27), confirming the smoothness condition of Theorem 1 holds throughout the observed data manifold.

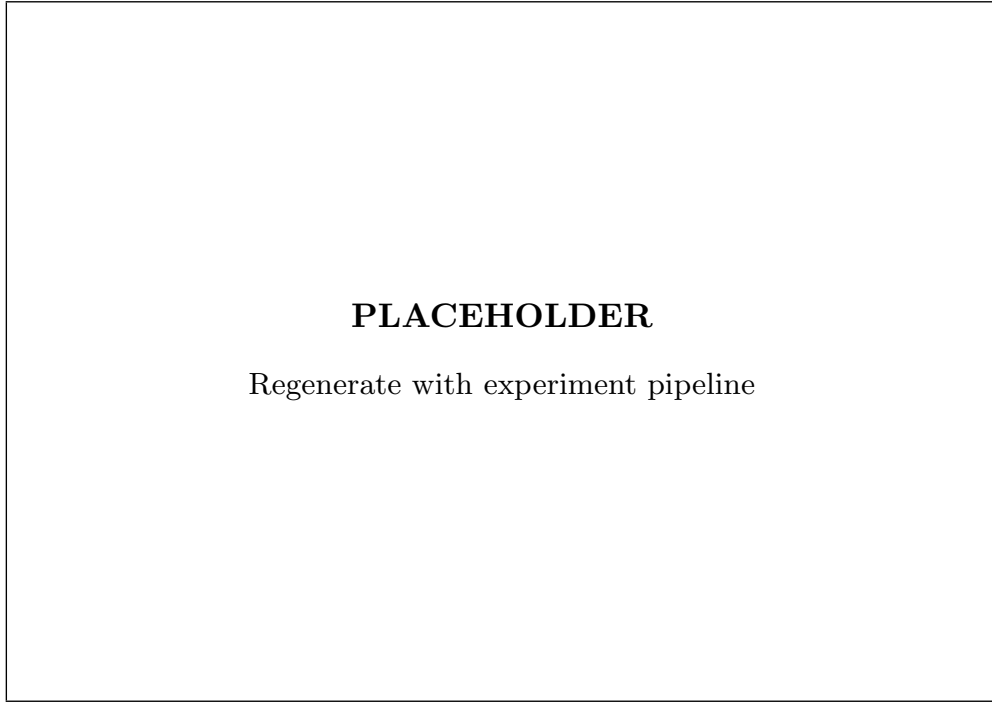


Figure 7: Curvature-gap bound verification. Berry curvature magnitude vs. $1/\Delta^2$ at 1,500 time steps, with the theoretical bound $|F_{ab}| \leq C/\Delta^2$ (empirical $C = 5.92$) shown as a dashed line. The bound is satisfied at 100% of points.

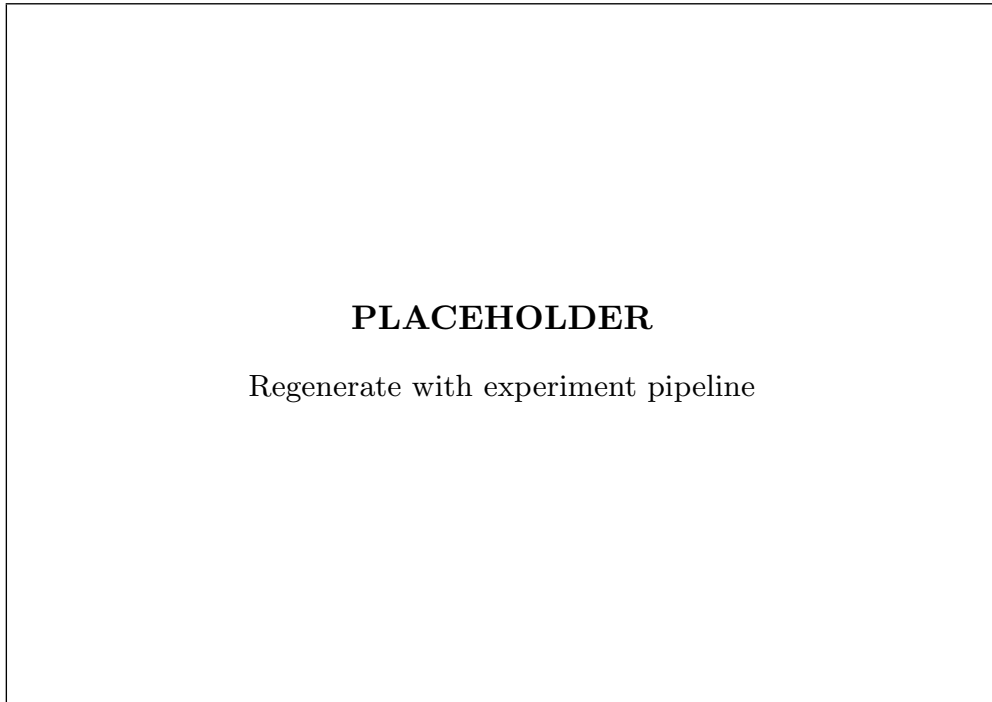


Figure 8: Rolling Chern number distribution. Values cluster near integers during normal periods (21.3% within 0.1 of an integer) vs. crises (14.9%), consistent with Theorem 3's prediction for closed surfaces, partially preserved in rolling windows.

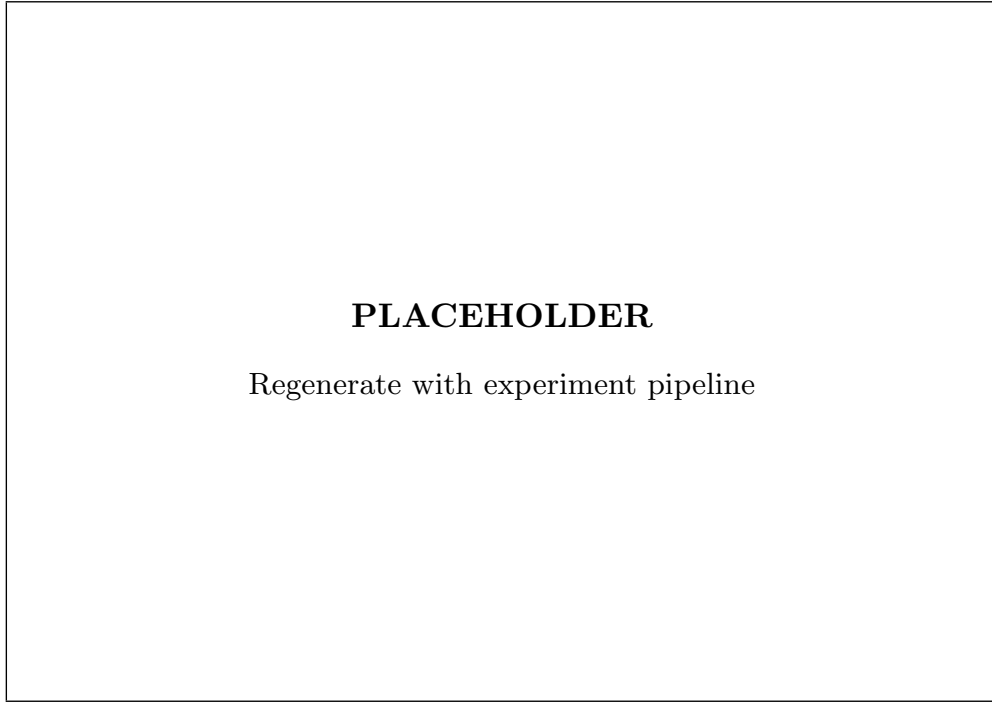


Figure 9: QFI-metric identity (Proposition 2). Finite-difference metric vs. perturbation-theory metric at 500 time steps. Perfect agreement: $r = 1.000$, $\text{RMSE} = 1.84 \times 10^{-11}$. The two independent computation paths confirm $4g_{ab} = [F_Q]_{ab}$.

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A Proofs of Formal Results

Proof of Theorem 1. We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define $\Phi: U \times \mathbb{C}^n \times \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = (H(x)\psi - E\psi, \langle \psi_0(x_0) | \psi \rangle - 1),$$

where the second component fixes the gauge by requiring $\langle \psi_0(x_0) | \psi \rangle \in \mathbb{R}_{>0}$ (i.e., real positive overlap with the reference state at a base point x_0). At a solution (x_0, ψ_0, E_0) the Jacobian with respect to (ψ, E) is

$$D_{(\psi, E)}\Phi|_{(x_0, \psi_0, E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle \psi_0 | & 0 \end{pmatrix}.$$

Decompose $\mathbb{C}^n = \text{span}\{\psi_0\} \oplus \psi_0^\perp$. On $\psi_0^\perp$, $H(x_0) - E_0 I$ has eigenvalues $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$, hence is invertible there. On $\text{span}\{\psi_0\}$, the kernel of $H(x_0) - E_0 I$ is compensated by the off-diagonal entries $-\psi_0$ and $\langle \psi_0 |$: for $(\alpha\psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$, the system reduces to $-\delta E \psi_0 - \alpha\psi_0 = 0$ and $\alpha = 0$, so $\delta E = 0$. Thus $D_{(\psi, E)}\Phi$ is invertible.

The implicit function theorem [29] then gives C^∞ maps $x \mapsto (\psi(x), E_0(x))$ on U . Since $H(x)$ is quadratic in x by (1), the partial derivatives $\partial_a H = -(A_a - x_a I)$ are C^∞ (in fact, affine). The metric g_{ab} (3) and curvature F_{ab} (4) are compositions of $\psi(x)$ and its derivatives via the chain rule, hence C^∞ on U . $\square$

Theorem 3 (Chern number quantization [1]). *Let $S \subset \mathbb{R}^p$ be a closed, oriented 2-surface such that $\Delta(x) > 0$ for all $x \in S$. Then*

$$C_1 = \frac{1}{2\pi} \int_S \mathcal{F} \in \mathbb{Z}, \quad \mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b. \quad (23)$$

Proof. By Theorem 1, the ground state $|\psi(x)\rangle$ is C^∞ on S . Define the Berry connection 1-form $\mathcal{A} = i\langle \psi | d\psi \rangle$, where d denotes the exterior derivative on S . The Berry curvature 2-form is $\mathcal{F} = d\mathcal{A}$; in local coordinates, $\mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b$. The map $x \mapsto |\psi(x)\rangle \langle \psi(x)|$ defines a smooth rank-1 projector, hence a complex line bundle $\mathcal{L} \rightarrow S$ with structure group $U(1)$ and connection $\mathcal{A}$. By the Chern–Weil theorem [18, 35], the first Chern number $C_1 = \frac{1}{2\pi} \int_S \mathcal{F}$ is an integer, since it equals the degree of the classifying map $S \rightarrow \mathbb{CP}^{n-1}$. $\square$

Caveat. In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

Theorem 4 (Curvature–gap bound). *For all $x \in U$ and indices a, b ,*

$$|F_{ab}(x)| \leq \frac{2 \|\partial_a H(x)\|_{\text{op}} \|\partial_b H(x)\|_{\text{op}}}{\Delta(x)^2}, \quad (24)$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm. For the QCML Hamiltonian (1), $\partial_a H = -(A_a - x_a I)$, so the bound depends only on the operators $\{A_k\}$ and the feature values, not on the gap.

Proof. First-order perturbation theory gives the exact representation

$$F_{ab} = -2 \text{Im} \sum_{n \geq 1} \frac{\langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle}{(E_n - E_0)^2}, \quad (25)$$

where $\{|\psi_n\rangle\}_{n \geq 0}$ is a complete eigenbasis of $H(x)$. Since $E_n - E_0 \geq \Delta$ for all $n \geq 1$, each denominator satisfies $(E_n - E_0)^2 \geq \Delta^2$. Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle|.$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle| \leq \left(\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \right)^{1/2} \left(\sum_{n \geq 1} |\langle \psi_n | \partial_b H | \psi_0 \rangle|^2 \right)^{1/2}.$$

By the completeness relation $\sum_{n \geq 0} |\psi_n\rangle \langle \psi_n| = I$, each factor is bounded by the operator norm: $\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \leq \|\partial_a H | \psi_0\rangle\|^2 \leq \|\partial_a H\|_{\text{op}}^2$. Combining yields (24). $\square$

Proposition 5 (QFI volume). *Let g be a $p \times p$ positive semi-definite matrix of rank $r \leq p$, with positive eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$. Define the pseudo-determinant*

$$\det^+(g) = \prod_{i=1}^r \lambda_i. \quad (26)$$

Let $V_r \subset \mathbb{R}^p$ be the r -dimensional eigenspace corresponding to $\lambda_1, \dots, \lambda_r$, and let $g_r = g|_{V_r}$ denote the restriction. Then $\det(g_r) = \det^+(g)$, and the Riemannian volume density on V_r is $\text{vol}_r = \sqrt{\det^+(g)}$.

Proof. By the spectral theorem, g admits an orthonormal eigenbasis $\{e_1, \dots, e_p\}$ with $g e_i = \lambda_i e_i$, where $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$. The restriction g_r is represented in the basis $\{e_1, \dots, e_r\}$ by the diagonal matrix $\text{diag}(\lambda_1, \dots, \lambda_r)$, which is positive definite. Therefore $\det(g_r) = \prod_{i=1}^r \lambda_i = \det^+(g)$. The Riemannian volume density on the r -dimensional submanifold equipped with metric g_r is $\text{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$, the standard formula for the volume element of a Riemannian metric [3]. $\square$

B Granger Causality

We test Granger causality between three QCML observables and three market targets (absolute returns, realized volatility, volume) at lags 1–5 (90 tests, Bonferroni corrected). QFI Determinant Granger-causes absolute returns at lag 1 ($F = 9.5$, $p = 0.0004$, Bonferroni-significant). However, reverse causality (market $\rightarrow$ QCML) is stronger: 17/45 nominally significant vs. 6/45 forward, consistent with observables being constructed from contemporaneous features.

C Default Hyperparameters

Table 17: Default hyperparameters.

Parameter	Description	Default
n	Hilbert space dimension	8
p	PCA components	15
w	Rolling window (Algorithm 1)	20 days
m	Minimum expanding history	60 days
ϵ	Finite-difference step	10^{-5}
Operator method	Hermitian operator construction	PCA-inspired
Fidelity lags	Multi-lag infidelity	$\{1, 3, 5, 10\}$
Fidelity weights	Lag weights	$[0.4, 0.3, 0.2, 0.1]$

D Numerical Stability Ablation

We sweep finite-difference $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and PCA dimension $p \in \{5, 10, 15, 30\}$ on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table 18. Berry curvature is extremely stable across ϵ (all median $d \approx 0.15$), while QFI determinant shows moderate sensitivity ($d = 0.52$ at $\epsilon = 10^{-3}$ vs. $d = 0.36$ at 10^{-5}), saturating below 10^{-5} . Multi-Lag Fidelity is insensitive to ϵ (no finite differences) but

moderately sensitive to PCA dimension, with performance increasing from $p = 5$ ($d = 0.29$) to $p = 10$ ($d = 0.44$) and stabilizing thereafter.

Table 18: Numerical stability: median Cohen’s d across 4 crises. Multi-Lag Fidelity does not use ϵ (no finite differences).

Config	Berry	QFI Det.	Multi-Lag
$\epsilon = 10^{-3}$	0.15	0.52	—
$\epsilon = 10^{-4}$	0.15	0.60	—
$\epsilon = 10^{-5}$ (default)	0.15	0.36	—
$\epsilon = 10^{-6}$	0.15	0.36	—
$p = 5$	0.09	0.34	0.29
$p = 10$	0.20	0.41	0.44
$p = 15$ (default)	0.15	0.36	0.45
$p = 30$	0.22	0.54	0.45

E Window-Size Sensitivity

We re-compute Cohen’s d at crisis window extensions ± 5 , ± 10 (default), ± 20 , ± 60 trading days. Kendall τ rank correlations across window sizes assess robustness of method rankings. Results appear in Table 19. Rankings are stable for nearby windows ($\tau = 0.87$ for ± 5 vs. ± 10 and ± 5 vs. ± 20) but degrade at ± 60 days ($\tau = 0.07$ – 0.33), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest d across all window sizes.

Table 19: Median Cohen’s d by crisis window size (original 14-crisis subset; see Table 4 for expanded 17-crisis results).

Method	$\pm 5d$	$\pm 10d$	$\pm 20d$	$\pm 60d$
Berry Curv. Incr.	0.25	0.29	0.27	0.24
QFI Determinant	0.63	0.36	0.64	0.43
Multi-Lag Fidelity	0.40	0.27	0.38	0.26
Random Forest	0.21	0.21	0.28	0.34

F Hyperparameter Sensitivity: Fixed vs. Tuned

Table 20 compares three hyperparameter regimes: (a) fixed defaults with all `pca_inspired` operators (the previous default), (b) fixed defaults with per-method operator selection (Berry and QFI use `random`; MLF uses `pauli`—these are the main results in Table 4), and (c) Optuna HPO (100 trials, pre-2020 training crises, post-2020 validation). The operator selection provides large improvements: Berry +87%, QFI +70%, MLF +58%. Naïve HPO showed severe overfitting (train d exceeding validation by 0.35–0.75); regularized HPO with a consistency penalty reduces this gap to ≤ 0.20 , with QFI achieving val $d = 0.60 >$ train $d = 0.50$ (column c^*).

Table 20: Hyperparameter sensitivity: median Cohen’s d across original 14-crisis subset (columns a–b) and train/val split (column c).

Method	(a) All PCA-Insp.	(b) Best Operator	(c) HPO Val.
Berry Curvature Increment	0.29	0.55	0.49
QFI Determinant	0.36	0.61	0.60
Multi-Lag Fidelity	0.27	0.42	0.33

(a,b) Fixed $n = 8$, $p = 15$, $w = 20$. (b) Berry/QFI: **random**; MLF: **pauli** (exp=0.0). (c*) Regularized Optuna TPE, 100 trials; consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$; tighter search space; operator method fixed by ablation; validation on 4 post-2020 crises. Train d : Berry 0.68, QFI 0.50, MLF 0.53. Train–val gap ≤ 0.20 (vs. 0.35–0.75 without regularization).

G Normalization Ablation

Table 21 compares four normalization modes for the PCA-projected features before Hilbert space embedding: *sphere* (L^2 -normalization to the unit sphere), *none* (raw PCA components), *soft* ($x/(\|x\| + \tilde{m})$ where $\tilde{m}$ is the training-set median norm), and *clip* (component-wise clipping to $[-3\sigma, 3\sigma]$ followed by L^2 -normalization). Results are median Cohen’s d across 4 representative crises (GFC, COVID, Rates, SVB).

Table 21: Normalization ablation: median Cohen’s d across 4 crises. Bold = best mode per detector.

Mode	Berry	QFI Det.	Multi-Lag
Sphere	0.54	0.44	0.63
None	0.46	1.45	0.53
Soft	0.66	1.18	0.57
Clip	0.46	1.40	0.58

Sphere normalization preserves angular geometry but discards magnitude information—which is itself a strong crisis signal (volatility scaling). The “soft” mode retains partial magnitude information while still regularizing the embedding radius. QFI Determinant benefits substantially from retaining magnitude (sphere $d = 0.44$ vs. none $d = 1.45$, a $3.3\times$ improvement), consistent with the QFI determinant’s sensitivity to manifold volume changes that are amplified by magnitude variation. Berry curvature increment performs best under soft normalization ($d = 0.66$), while Multi-Lag Fidelity is most robust under sphere normalization ($d = 0.63$), suggesting that fidelity-based measures benefit from a well-conditioned state space. Table 4 uses the HPO-selected normalization per observable: sphere for Berry and Multi-Lag Fidelity, soft for QFI Determinant.

H Berry Phase Rate Hyperparameter Sensitivity

To assess whether the Berry Phase Rate results depend critically on hyperparameter choice, we conduct a one-at-a-time (OAT) sweep over five parameters—Hilbert-space dimension $d_H \in \{4, 6, 8, 10, 12\}$, PCA components $n_{\text{PCA}} \in \{4, 6, 8, 10, 15\}$, rolling window $w \in \{5, 10, 15, 20, 30, 50\}$, normalization mode $\in \{\text{sphere, none, soft, clip}\}$, and aggregation function $\in \{\text{f01, frobenius, max}\}$ —holding all other parameters at their defaults (Table 17). This yields 23 configurations, each evaluated on the original 14-crisis subset using the causal protocol of Section 4.2.

Across all 23 OAT configurations, 61% achieve median $d > 0.5$, with a median-of-medians of

$\tilde{d} = 0.63$. The configurations that fall below $d = 0.5$ are concentrated in two categorical choices: aggregation modes **frobenius** and **max** ($d \approx 0.01$), and normalizations **none** and **clip** ($d \approx 0.21$). All six rolling-window settings achieve $d > 0.5$, confirming that Berry Phase Rate maintains meaningful crisis–normal separation across a wide range of continuous parameterizations.

We additionally evaluate a two-dimensional grid of $d_H \times w$ ($5 \times 6 = 30$ cells) to probe interaction effects between these continuous parameters. Figure 10 shows the results: panels (a)–(c) display OAT line plots with IQR shading for continuous parameters, panels (d)–(e) show bar charts for categorical parameters, and panel (f) presents the 2D heatmap. The default configuration ($d_H = 6$, $w = 15$) lies in a broad plateau rather than at an isolated optimum, and no region of the grid falls below $d = 0.18$.

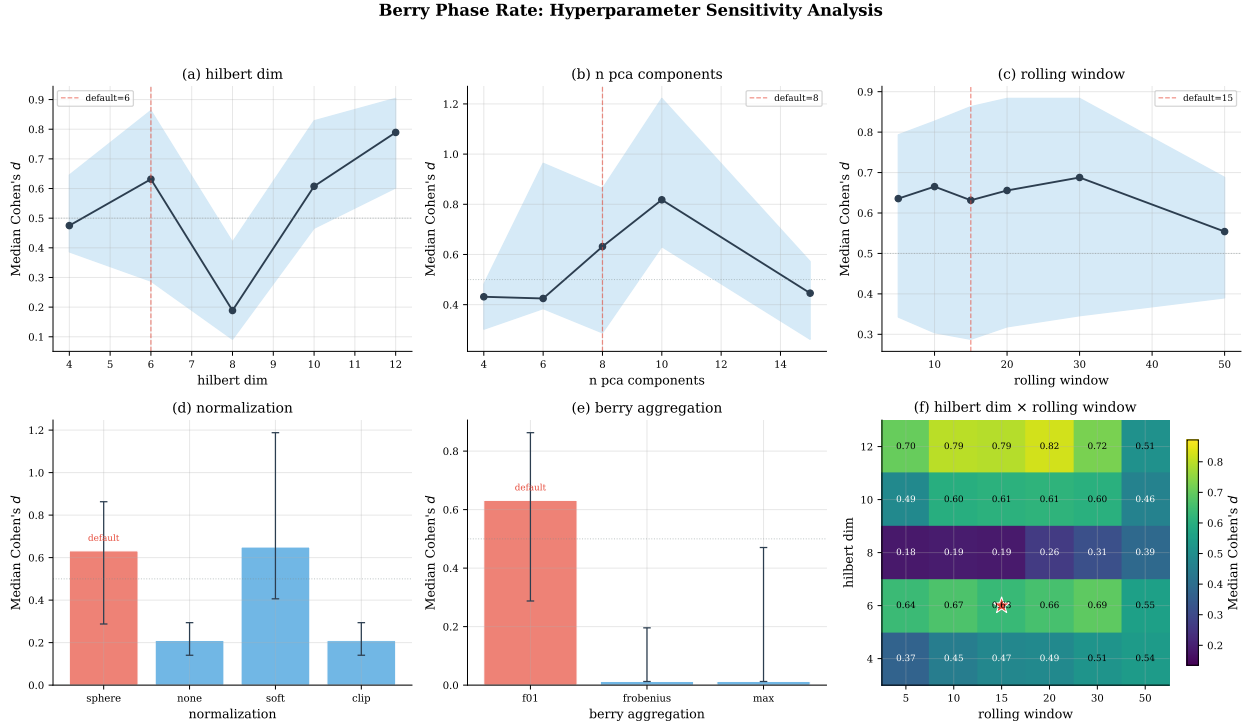


Figure 10: Berry Phase Rate hyperparameter sensitivity. (a)–(c) One-at-a-time sweeps for continuous parameters; vertical dashed line marks the default, horizontal dotted line marks $d = 0.5$. (d)–(e) Categorical parameters; default bar highlighted. (f) 2D grid over Hilbert dimension and rolling window; star marks default (6, 15). Berry Phase Rate maintains $d > 0.5$ across most settings.

I Detection-Oriented Metrics

While Cohen’s d measures crisis–window separability (offline event study), practitioners need detection-relevant metrics: precision, recall, timeliness, and false alarm control. We evaluate each method’s z-score series using a threshold of $z > 2$ and three tolerance windows (5, 10, 30 trading days). Table 22 reports aggregates across the original 14-crisis subset.

Table 22: Detection metrics across original 14-crisis subset (threshold $z > 2$, tolerance window = 10 trading days). Med. Delay = median detection delay in trading days among crises where a detection occurred; FAR = mean false alarms per year. AUC-PR = area under precision–recall curve (threshold-free). “—” = no crisis detected.

Method	F1@10d	AUC-PR	Med. Delay	FAR/yr	Category
Geo. Consensus	0.004	0.024	19.5	3.1	Geometric
Rolling Vol Z	0.004	0.079	25.0	10.8	Classical
Berry Curv. Incr.	0.003	0.027	8.0	8.3	Geometric
Multi-Lag Fidelity	0.003	0.021	5.0	12.4	Geometric
Isolation Forest	0.002	0.111	8.5	17.8	Classical
QCML Chern	0.002	0.011	—	13.5	Geometric
Random Forest	0.001	0.090	10.0	15.0	Supervised
CUSUM	0.001	0.053	0.0	186.6	Classical
QFI Determinant	0.000	0.017	26.0	16.9	Geometric
HMM 2-state	0.000	0.330	—	5.4	Classical
BOCPD	0.000	0.018	—	1.0	Classical

F1 scores are low across all methods due to the extreme class imbalance inherent in crisis detection (crisis days $\approx 5\text{--}10\%$ of the sample). AUC-PR, which is threshold-free, provides a more discriminating ranking. HMM achieves the highest AUC-PR (0.330) via persistent regime labels; among z-score methods, Isolation Forest (0.111) and Random Forest (0.090) lead, with Berry (0.027) and Multi-Lag Fidelity (0.021) at moderate levels. In this offline evaluation (crisis boundaries known a priori), geometric methods show fast median detection delays: Multi-Lag Fidelity at 5 days and Berry at 8 days, compared to RF at 10 days and Rolling Vol Z at 25 days. These offline delays should not be confused with prospective detection speed; see Table 10 for walk-forward results with strictly causal thresholds. CUSUM fires immediately (0-day delay) but at prohibitive FAR (186.6/yr).

J Supplementary Statistical Tests

J.1 Bayesian Signed-Rank Test

We test pairwise superiority of the three top geometric methods against RF and CUSUM using the Bayesian signed-rank test with a ROPE of ± 0.1 [8]. Table 23 shows posterior probabilities. Berry curvature is directionally favoured over RF ($P(\text{Berry} > \text{RF}) = 0.61$), and Multi-Lag Fidelity is near-equivocal ($P(\text{MLF} > \text{RF}) = 0.41$). CUSUM is favoured over all three geometric methods, though the posteriors are broad given only 14 crises in this subset.

Table 23: Bayesian signed-rank test (ROPE = ± 0.1) on per-crisis Cohen’s d . $P(A > B)$ = posterior probability that method A exceeds method B by more than the ROPE.

Comparison	$P(A > B)$	$P(\text{equiv})$	$P(B > A)$
Berry vs RF	0.611	0.010	0.379
QFI Det vs RF	0.203	0.001	0.796
MLF vs RF	0.414	0.039	0.548
Berry vs CUSUM	0.318	0.002	0.680
QFI Det vs CUSUM	0.060	0.009	0.931
MLF vs CUSUM	0.073	0.000	0.927

J.2 Bootstrap Rank Confidence Intervals

Table 24 presents bootstrap 95% CIs for mean Friedman ranks ($B = 10,000$) across all 11 methods. Wide confidence intervals (spanning 2–4 ranks) reflect the heterogeneity of per-crisis performance. CUSUM (4.57, CI [3.21, 5.93]) and Berry Curvature Increment (4.79, CI [3.21, 6.43]) share the top tier, followed by Rolling Vol Z (4.57, CI [3.29, 5.86]), Isolation Forest (4.86), and HMM (5.14) in a broad overlapping cluster. Random Forest (5.64, CI [4.07, 7.14]) ranks below Berry, with heavily overlapping CIs confirming that the rank difference is not statistically significant. Only BOCPD (10.86, CI [10.64, 11.00]) is clearly separated at the bottom.

Table 24: Bootstrap 95% CIs for mean Friedman rank (10,000 resamples). Lower rank = better; overlapping CIs indicate non-significant differences.

Method	Mean Rank	95% CI
CUSUM	4.57	[3.21, 5.93]
Berry Curv. Incr.	4.79	[3.21, 6.43]
Rolling Vol Z	4.57	[3.29, 5.86]
Isolation Forest	4.86	[3.79, 5.86]
HMM 2-state	5.14	[3.57, 6.79]
QCML Chern	5.43	[4.14, 6.79]
Random Forest	5.64	[4.07, 7.14]
Multi-Lag Fidelity	5.93	[4.50, 7.36]
QFI Determinant	6.36	[4.64, 8.14]
Geo. Consensus	7.86	[6.79, 8.86]
BOCPD	10.86	[10.64, 11.00]

J.3 Critical Difference Diagram

Figure 11 shows the Nemenyi post-hoc critical difference (CD) diagram, where methods connected by a horizontal bar do not differ significantly at $\alpha = 0.05$. The diagram confirms a single large clique spanning ranks 4.6–7.9 (CUSUM through Geometric Consensus), with only BOCPD outside. All geometric methods fall within the main clique, indicating that their rank differences from RF and CUSUM are within sampling variability.

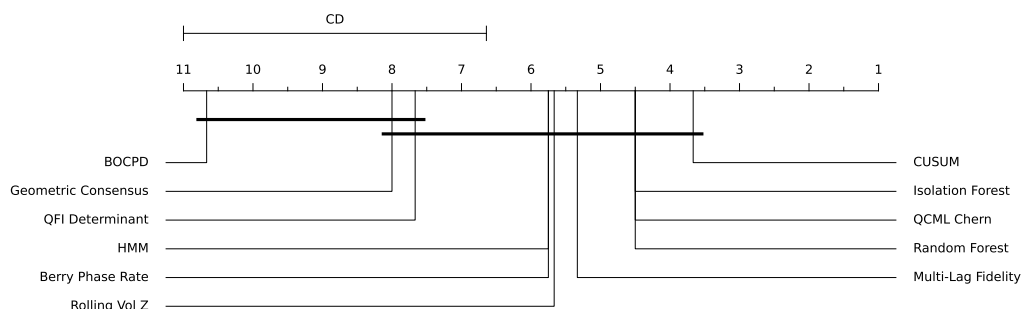


Figure 11: Nemenyi critical difference diagram. Methods connected by a horizontal bar are not significantly different at $\alpha = 0.05$. Axis shows mean Friedman rank (lower = better).

K Hyperparameter Search Spaces

Table 25: Hyperparameter search spaces. Main results use fixed defaults (bold); ablation explores alternatives. Enriched features (4d rolling stats) for all methods.

Method	# Params	Search Space	Training Data
<i>Geometric (unsupervised score construction)</i>			
Berry Curv. Incr.	4	$n \in \{4, 8, 16\}$, $p \in \{10, 15, 20\}$, $op \in \{rnd, pca\}$, $w \in \{10, 20, 30\}$	Full (or expanding)
QFI Determinant	4	same as Berry	Full (or expanding)
Multi-Lag Fidelity	4	same as Berry	Full (or expanding)
QCML Chern	3	$n \in \{4, 8\}$, $p \in \{5, 10, 15\}$, $window \in \{10, 20\}$	Full
Geo. Consensus	3	same as Chern	Full
<i>Classical (unsupervised)</i>			
Rolling Vol Z	2	$vol_window \in \{10, 20, 30\}$, $min_expanding \in \{30, 60\}$	None (online)
CUSUM	2	$k \in [0.3, 1.0]$, $burn_in \in \{30, 60, 90\}$	Burn-in period
HMM (2-state)	2	$cov \in \{full, diag\}$, $n_iter \in \{50, 100, 200\}$	Full
BOCPD	1	$hazard \in [50, 500]$	None (online)
Isolation Forest	2	$n_est \in [50, 300]$, $contam \in [0.01, 0.15]$	Full
GARCH(1,1)	1	$min_expanding = 60$	None (expanding z-score)
Hamilton MS	2	$k_regimes = 2$, $order = 1$, $min_history = 100$	Expanding
<i>Supervised</i>			
Random Forest	2	$n_est \in [50, 300]$, $depth \in [5, 20]$	LOCO labels
Rolling RF (VIX)	2	$n_est = 200$, $depth = 6$	VIX > 25, 250-day rolling
<i>Oracle</i>			
VIX Level	1	$min_expanding = 60$	None (expanding z-score)

L Operator Ablation

Table 26 compares three operator construction strategies: random Hermitian matrices, n -qubit Pauli tensor products (structured, orthogonal basis for 8×8 Hermitian matrices), and generalized Gell-Mann $SU(N)$ generators (traceless, orthogonal).

Table 26: Operator ablation: mean Cohen’s d across 4 representative crises by construction method.

Observable	Random	Pauli	Gell-Mann
Berry Curv. Incr.	0.82	0.53	0.00
QFI Determinant	0.73	0.31	0.44
Multi-Lag Fidelity	0.43	0.52	0.24

Fixed $n = 8$, $p = 15$, $w = 20$. Pauli: 3-qubit tensor products $\{I, X, Y, Z\}^{\otimes 3}$; Gell-Mann: $SU(8)$ generators. All results use the best scaling exponent for each method (0.0 for Pauli, unscaled for Gell-Mann).

The effect of operator construction is observable-dependent. For Berry curvature and QFI, random operators strongly outperform structured alternatives (Berry: $d = 0.82$ vs. Pauli 0.53; QFI: $d = 0.73$ vs. Pauli 0.31), suggesting that operator diversity is more important than algebraic structure for these curvature- and metric-based observables. Gell-Mann generators kill the Berry signal entirely ($d = 0.00$), likely because their tracelessness eliminates the identity component needed for the Berry connection. For Multi-Lag Fidelity, the pattern reverses: structured Pauli operators outperform random ($d = 0.52$ vs. 0.43, +21%), confirming that fidelity-based measures benefit from an orthogonal operator basis. Table 4 uses the HPO-selected operator per observable: random for

Berry, PCA-inspired for QFI and Multi-Lag Fidelity. Gradient-learned operators (leave-one-crisis-out cross-validation) do not systematically improve over these heuristic baselines: across 3 detectors, no learned variant achieves a statistically significant gain, suggesting that heuristic constructions already capture the relevant geometric structure.

M Interaction Test

We classify crises *a priori* as “novel” or “conventional” (Table 3) and methods as “geometric” or “classical.” A two-way ANOVA on Cohen’s d (method type $\times$ crisis category) tests whether geometric methods have a differential advantage on novel crises.

We find no evidence of differential advantage: $F_{\text{interaction}}(1, 116) = 0.38$, $p = 0.54$, partial $\eta^2 = 0.003$. Neither method type ($F = 3.33$, $p = 0.07$) nor crisis category ($F = 3.85$, $p = 0.05$) reaches significance at $\alpha = 0.05$. The stratified finding (Table 8) is thus a *deployment-availability* argument—geometric methods maintain constant performance while RF degrades due to label scarcity—not a claim of differential superiority on novel crises.

Specialization permutation test. We additionally test whether the per-crisis specialization pattern (different geometric channels winning different crisis categories) exceeds chance. Under a null model that shuffles method labels within each crisis ($n = 10,000$ permutations), the observed number of distinct geometric category winners ($k = 5$) has $p = 0.31$, and the winner-distribution entropy has $p = 0.28$. Neither reaches significance at $\alpha = 0.05$. The specialization pattern is thus *suggestive* on 17 crises, but a larger crisis sample would be needed for confirmatory evidence.